COURSE
OVERVIEW

FALL PROTECTION PLAN DEVELOPMENT

The Fall Protection Plan Development course provides learners with knowledge to develop compliant fall protection plans for work-at-height activities. The course focuses on hazard identification, risk assessment, and control selection.

Learners are introduced to legal requirements, documentation, system selection, and rescue planning. Emphasis is placed on structured planning.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Explain legal requirements of fall protection planning
- Identify fall hazards
- Conduct risk assessments
- Select suitable fall protection systems
- Compile site-specific fall protection plans
- Include training and rescue requirements

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GRAVITY VALUES

R — RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

I — INTEGRITY

Conducting day-to-day business with integrity at all times.

S — SOLUTIONS **DRIVEN**

Being solution driven.

Q — QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

Before any work begins, the site must be safe.
Use this simple checklist

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

01

Rescue Plan

Have a rescue plan ready.
A rescue kit and first aid kit must be on site.

02

Work in Pairs

There must be at least two workers on site.
If something goes wrong, one can help or call for help.

03

Check Equipment

All tools and gear must be safe and working before use.

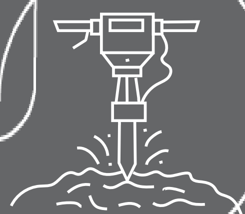
04

Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.

05



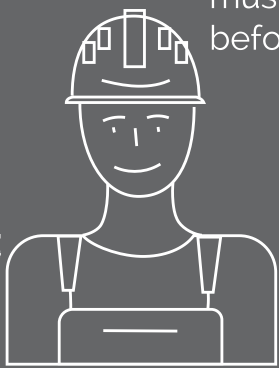
SITE ESTABLISHMENT

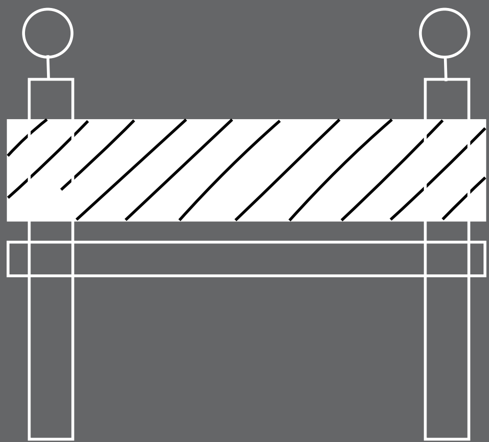
SITE ESTABLISHMENT

06

Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.





07

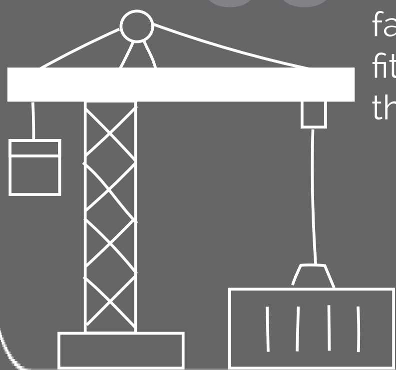
Do a Risk Assessment

Check what dangers there are. All workers must understand the risks before starting work.

08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

- **Be healthy and ready** before starting work.
- **Only do jobs you are trained** and allowed to do.
- **Never work under the influence** of alcohol or drugs.
- **Always wear your safety gear** (PPE, harness if working at height).
- **Use harnesses correctly** and stay connected at heights.
- **Never work alone** and always follow safety steps.

SITE SAFETY

- **Keep people away** from areas below when working above.
- **Don't walk under lifted loads** or stand in dangerous areas.
- **Only trained people** can do electrical, underground, or confined space work.
- **Follow road rules.** Wear helmets. No passengers on open trucks..
- **Don't drive tired** or distracted.
- **Stop unsafe work!** Everyone has the right to speak up.

GRAVITY


CAMP.

A. ABBREVIATIONS AND DEFINITIONS

1. Abbreviations and Acronyms

- **CE** – Conformité Européene
- **CEIM** – Climbing Equipment Inspection and Management
- **COF** – Certificate of Fitness
- **DSTI** – Daily Safe Task Instruction
- **EN** – European Norm
- **FAS** – Fall Arrest System
- **FPP** – Fall Protection Plan
- **HIRA** – Hazard Identification and Risk Assessment
- **HSE** – Health, Safety and Environment
- **ICOP** – International Code of Practice
- **IRATA** – Industrial Rope Access Trade Association
- **MBS** – Minimum Breaking Strain
- **MEWP** – Mobile Elevated Work Platform
- **MFS** – Minimum Free Space
- **NQF** – National Qualifications Framework
- **OEM** – Original Equipment Manufacturer
- **OHP** – Occupational Health Practitioner (registered)
- **OHS Act** – Occupational Health and Safety Act 85 of 1993
- **OMP** – Occupational Medical Professional (registered)
- **PPE** – Personal Protective Equipment (mainly focused on work at height)
- **PTO** – Planned Task Observation
- **RA** – Rope Access
- **RFA** – Radio Frequency Awareness
- **SETA** – Sector Education and Training Authority
- **SOP** – Standard Operating Procedure
- **SWL** – Safe Working Load
- **TACS** – Training Assessment and Certification Scheme

2. Definitions

1. **Anchor Point** – An attachment point at an anchor for anchor lines or persons (EN 795).

2. **Competent Person** – An individual who, by way of training/qualification, skill and/or experience, is knowledgeable of applicable regulations and standards, can identify workplace hazards relating to the specific operation, is designated in writing by the employer, and has authority to take appropriate actions.

3. **Design** – Includes drawings, calculations, design details and specifications relating to a structure.

4. **Designer** –

a. A person who prepares a design;
b. a person who checks and approves a design;
c. a person who arranges for any person at work under their control (e.g. an employee) to prepare a design;
d. an architect or engineer contributing to or having overall responsibility for the design;
e. a building services engineer designing details for a fixed plant;
f. a surveyor specifying articles or drawing up specifications;
g. a contractor carrying out design work as part of a design and building project;
h. any competent person designing falsework, including its components, form work and support work;
i. an interior designer, shopfitter or landscape architect.

5. **Energy-Absorbing Lanyard** – The connecting element or component of a fall arrest system. A lanyard may be of synthetic-fibre rope or webbing. It must have an energy-absorbing component incorporated into it, which will keep the impact force below 6 kN shock force in the event of a fall.

6. **Equipment Controller** – A person appointed, in writing, to perform three-monthly inspection and management on all work-at-height equipment within the organisation, as per the fall protection plan.

7. **Exclusion Zone –** An area into which entry is forbidden, where tools or equipment can fall off or onto an object or person in the surrounding area.

8. **Fall Arrest** – A form of fall protection which involves safely stopping a person in the event of a fall.

9. **Fall Arrest Equipment** – Equipment used to arrest a person in a fall, including personal equipment such as a body harness, lanyards, deceleration devices, lifelines or similar equipment. This excludes single belts. The equipment is designed to stop (arrest) a person's fall in the safest and shortest distance possible.

10. **Fall Arrest and Rescue Technician** – A person who can perform fall arrest manoeuvres and a basic fall arrest rescue, and who is in possession of a valid Fall Arrest and Basic Rescue certificate.

11. **Fall Arrest System (FAS)** – An assembly of components joined together so that, when connected to a suitable anchor point with sufficient clearance from the ground or other obstacles, it operates as a complete arrangement of equipment able to arrest a fall. This may include the safe use of a climbing helmet with chin strap (EN 397), full body harnesses (EN 361, EN 358, EN 813), energy absorbers (EN 355), vertical lifelines (EN 353, EN 795), horizontal lifelines (EN 353-1, EN 353-2, EN 795), retractable lifelines (EN 360), using a work positioning lanyard (EN 358) as body support, etc.

12. **Fall Prevention Equipment** – Equipment used to prevent persons from falling from a fall risk position, including personal equipment such as a body harness, lanyards and lifelines, and physical equipment such as guardrails, screens, barricades, anchorages or similar equipment.

13. **Fall Protection Plan (FPP)** – A documented plan of all risks relating to working from a fall risk position, considering the nature of the work undertaken and setting out the procedures and methods to be applied to eliminate the risk. It must include a rescue plan and procedure.

14. **Fall Risk** – Any potential exposure to a risk of falling (items falling on workers, workers falling from an elevated position or workers dropping items onto other workers). Work cannot continue onsite unless the fall risks have been mitigated, reduced and controlled to acceptable levels.

15. **Fixed Barricade** – Barricading provided to prevent persons from entering areas where there is a fall risk, a slipping risk, potential falling object, open trenches or holes, unprotected edges, etc.

16. **Full Body Harness** – Body support primarily for fall arrest purposes, i.e. a component of a fall arrest system. A harness may comprise of straps, fittings, buckles or other elements suitably designed to support the body and provide attachment points for the working line and safety line for rope access work.

17. **Hazard** – A source of or exposure to danger (a source which may cause injury or damage to persons).

18. **Hazard Identification** – The identification and documentation of existing or expected hazards to the health and safety of persons, which are normally associated with the type of construction work being executed or to be executed.

19. **Health, Safety and Environmental Plan (HSE Plan)** – Dedicated health and safety management system that ensure appropriate standards of health, safety and well-being are maintained by employees.

20. **Inclement Weather** – Any weather condition that can result in an unsafe work environment. This will be specific to the task and rescue scenario.

21. **Medical Fitness** – Being declared medically fit to work safely from a fall risk position or a similar environment, and being in possession of a valid medical certificate of fitness.

22. **Medical Certificate of Fitness (COF)** – A certificate valid for one year, issued by a medical practitioner. This certificate may become temporarily or permanently invalid following consultation with a medical practitioner if a medical or mental condition arises or is diagnosed within the 12-month validity period. Also known as a Medical and Psychological Certificate of Fitness.

23. **Minimum Free Space (MFS)** – The direct vertical area from the point of anchorage to the ground. Fall arrest systems are designed to limit the force on the body during a fall. This is achieved by absorbing the energy generated in the fall by applying an arresting force to the faller over a distance, i.e. the "arrest distance". In order to prevent the possibility of a collision, there must be sufficient free space under the user for the fall to be arrested, i.e. the free space must be greater than the arrest distance. Free space ensures that the path of the fall is free from obstacles.

24. **Mobile Elevated Work Platform (MEWP**) – A mobile raised platform used to provide temporary access for technicians or equipment to otherwise inaccessible areas at height.

25. **Night Work** – Work conducted in the period between sunset and sunrise. This period is considered to start 15 minutes before sunset and end 15 minutes before sunrise.

26. **Permanent Barricade** – Barricading that is permanently fixed, that is used to prevent falling or provide support when ascending or descending stairways, such as handrails.

27. **Portable Ladder** – A temporary structure that can be moved or extended to a desired length. Portable ladders consist of side rails and rungs. There are various types of portable ladders that can be made of various types of materials.

28. **Pre-Task Risk Assessment** – A risk assessment that is completed by the site supervisor at the worksite before work may start.

29. **Risk Rating** – A rating indicating the probability or likelihood that a hazard can result in injury to people or damage to property.

30. **Risk Assessment** – A process of determining all risks associated with all hazards at a specific construction site, in order to identify the steps needed to be taken to eliminate, mitigate and/or control such hazards.

31. **Rope Access (RA)** – A climbing technique using ropes, normally incorporating two separately secured systems, one as a means of access and the other as backup security. The ropes are used with a harness and in combination with other devices, facilitating movement to and from the place of work and work positioning.

32. **Rope Access Basic Operative -** A person who can perform rope access manoeuvres and a basic rope access rescue, and who is in possession of a valid Rope Access Level 1 certificate.

33. **Rope Access Intermediate Operative** – A person who can perform rope access manoeuvres, supervision and advanced rescues on a simple worksite, and who is in possession of a valid Rope Access Level 2 certificate.

34. **Rope Access Advanced Operative** – A person who can perform rope access manoeuvres and a set of advanced rope access rescues on a complex worksite, and who is in possession of a valid Rope Access Level 3 certificate. A rope access level 3 supervisor can supervise work on both a simple and complex worksite.

35. **Scaffolding** – A temporary structure used to access a worksite. It is used to support technicians and equipment when building, repairing or cleaning buildings.

36. **Site Supervisor** – A person, appointed in writing, who takes full responsibility of all work-at-height activities onsite and ensures that all work-at-height processes are conducted in the correct manner.

37. **Suspended Platform** – A working platform suspended from supports by one or more separate ropes from each support, also known as a suspended access platform, temporary suspended platform (TSP), building maintenance unit (BMU), etc.

38. **Toolbox Talks** – Talks which communicate the daily hazards and risks onsite to the workers, and which provide an opportunity to reinforce the safe-working procedures and necessary personal protective equipment for the specific daily task. It is the most practical step in bridging the gap between paperwork and the actual worker.

39. **Working Platform** – A platform with the purpose to support the combined imposed loads of the workers' materials and plant. The number of working platforms permitted in simultaneous usage on scaffolding frameworks is limited.

40. **Work** – Any task, observation or inspection performed by an employee.

41. **Work at Height** – Any work performed above or below a stable work surface, or where a person puts themself in a position where they are exposed to a fall hazard/risk (i.e. where they can fall from or into an area that will likely result in an injury).

42. **Work Positioning** – A technique that enables a person to work supported in tension or suspension by personal protective equipment in such a way that a fall from height is prevented or restricted.

43. **Work Restraint** – A technique whereby a person is prevented by means of a harness and other devices from reaching zones where the risk of a fall from height exists.

B. INTRODUCTION TO FALL PROTECTION PLANNING

1. Planning

When you go on vacation, what process do you follow? Do you just jump in your car and drive without knowing where you are heading, or do you sit with your family and plan where you want to go, how long you will be there, what you are going to eat, etc.?

Proper planning prevents poor performance.

We plan for work at height for the same reason we plan for anything else: Planning prepares us for what we may encounter, including the worst-case scenarios. Without proper planning, we will not be able to ensure the safety of our workers in an environment where a mistake could be the difference between life and death.

This means we must have a detailed plan and methodologies for everything we will do to ensure the safety of our workers.

In order to do this, we need to understand what work at height encompasses so that we know what it looks like and what we are planning for. Let's start by considering different access methods.

2. Different Work and Access Methods

i. Access Control Methods

When we look at our hierarchy of control, our first goal is to eliminate the hazard completely. While working at height, our primary hazard is falling. Whether it be people or objects, falling is what we are trying to prevent.

Naturally, we first try to avoid a situation where a worker is exposed to the possibility of falling. We try to steer clear of the hazard altogether, meaning work is only done when there is no fall risk.

If that is not possible, we try to prevent access to an area where a fall can occur, by means of barricading and awareness training to keep workers out of that area. This is the first basic step.

ii. Work Restraint Systems

If we cannot avoid an area where a fall is possible, we move to our next best practice, called work restraint. Work restraint is applied in areas where workers must work close to a position where a fall might occur, but where they do not have to work in a fall risk position. The most common example of this type of exposure is on top of a roof, where a worker might move within two metres of the edge of the roof, but they will not need to move beyond that edge.

Unfortunately, work restraint is not always possible. If we look at a telecoms structure, for example, there is no way we can work on the antenna level without being in a position where a fall can occur.

iii. Scaffolding and MEWPs

When it comes to accessing an area where work will take place, the ideal access method would be the ability to walk up to that area, with no additional requirements. This would mean the use of walkways, ramps or stairs.

If we cannot access an area of work with just walkways, ramps or stairs, because it is either too high or too steep, the next method would be to construct a structure to access that area, such as scaffolding.

However, scaffolding can be very time-consuming to erect and, therefore, alternative methods have to be considered. This is where MEWPs come in.

MEWPs enable you to reach up to 45 m from ground level without the strain of climbing to the level you want to work on.

Parallel to MEWPs, you could also consider the use of a man basket, especially if there is no space to park a MEWP. Man baskets typically work from the top down and would not require you to use space on the ground level. It also allows workers to descend from the tops of buildings, and as a result, workers can reach much higher areas than they can with MEWPs.

These two methods can be quite expensive, though, and have certain limitations in mobility. With rope access, however, the worker can reach the same areas as the man basket, without the basket. Rope access is also much cheaper, and the same equipment can easily be transported and used on a large variety of sites, without any additional needs.

iv. Fall Arrest

All the abovementioned access methods have the same basic hazard – falling. Since we cannot eliminate the risk of falling completely, we need to look at making falling safe.

People have been falling safely for decades. Take, for example, activities like skydiving, base jumping, etc. All these activities work on the principle that the fall is arrested or slowed down before the point where it will lead to injury – this is called fall arrest. With proper training and equipment, workers can safely work 50 m in the air, for example.

All this is made possible by what we call a shock absorber. The shock absorber has a lanyard that gradually slows the fall of a worker to prevent shock load.

Fall arrest is currently the most popular solution for safely working at heights, but this method is dependent on being able to climb to the point where you want to work. An example of this is a fixed ladder to access a roof – you have a place to hold on with your hands and a place to stand with your feet.

With fall arrest, we focus on a few critical components. First, we look at maintaining a safe fall factor. Then, we consider how the fall factor impacts MFS, how to use vertical and horizontal lifelines (fixed and temporary), and how to safely use retractable lanyards (including their advantages and disadvantages).

But what happens when there is no place to hold on to or stand on?

v. Portable Ladders

When it comes to lower-level heights, humankind has invented portable attachment access points and ways to climb, such as portable ladders and pole climbing shoes. And though these methods are extremely effective, it is typically restricted to heights of nine metres.

Portable ladders account for the majority of falls from height, due to the general assumption that people know how to use them correctly. Falls from ladders are generally caused when:

- wrong types of ladders are used for the task,
- ladders are not safe for use,
- ladders have not been manufactured to the relevant standard,
- workers have not been trained on how to use ladders or
- ladders are transported incorrectly.

It should be made clear that portable ladders add significant risk to a worksite, and these risks should be mitigated by ensuring that all work carried out on portable ladders is done so by trained and competent technicians. This requirement must be included in the training matrix.

vi. Scaffolding

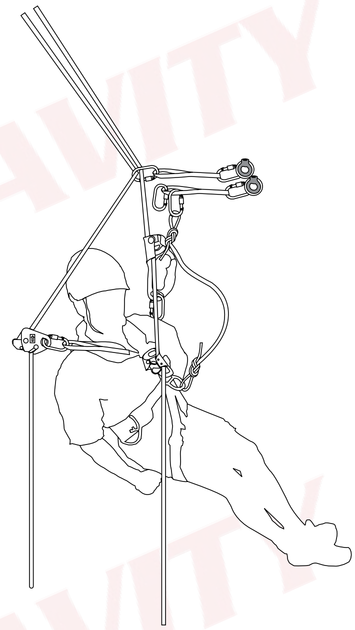
Currently, most companies consider scaffolding for work that must be performed at height. With scaffolding, you can build a temporary access structure for continuous long-term use.

Scaffolding as a solution has been used as far back as the construction of the pyramids and maybe even further back. It is a great solution, but it takes time to build and can be quite costly, especially if the structure is hired per day and the project will take weeks.

You have to ask yourself, for example, "Is it practical to spend two weeks to erect a 50 m scaffold next to a building for the sake of installing nine cables, when the actual installation would only take four hours to do?"

vii. Rope Access

This brings us to our next access method, called rope access. Rope access is the access method used to ascend or descend on the side of or inside a structure if there is no place to stand or to grab onto with your hands. There are two ropes going up or down, connected to two separate anchor points. These ropes are called your main line and backup line, respectively. Rope access can be applied to various scenarios to enable the user to access what would appear to be inaccessible places.



It is worth noting that there is a difference between the equipment used for rope access and the equipment used for fall arrest. Basic fall arrest equipment will not be sufficient for rope access work.

Depending on the complexity of the work being done, different levels of supervision will be required. A basic operative, also known as a level 1 technician, may never operate without the supervision of an intermediate or advanced operative. An intermediate operative, also known as a level 2 technician, may supervise simple sites. A simple site is any site where there will only be up-and-down movement on the same ropes and no side-to-side movement.

An advanced operative, otherwise known as a level 3 technician, can supervise all other work. The reason for this is the assumption that an advanced operative has more experience because of the prerequisites for the advanced operative course. Before a technician may attend the advanced operative course, they need a thousand hours as a basic operative, under supervision of an intermediate or advanced operative, and then another thousand hours as an intermediate operative. In practice, this takes a few years to achieve, and includes working on a variety of different sites with most types of equipment. Because of this, an advanced operative is also deemed to be a competent gear inspector.

viii. Rescue

For each of the access methods, we also need to consider that because someone is working in a fall risk position, there is always the possibility that they may fall. Though this fall is less likely to be fatal because of necessary equipment and training, a person may still need to be rescued if a fall does occur.

For that reason, rescue is taught alongside each of the different access methods. With work restraint as well as pole climber and ladder user courses, we teach rig-for-rescue in case a person does manage to fall over the edge or off the structure where work is performed. Rig-for-rescue allows the rescuer to simply lower the fallen worker to the nearest platform or all the way to the ground. In fall arrest, we teach the basic lowering rescue, also known as the ratchet rescue.

In the basic operative course, we teach a simple same-rope rescue from a descender, and in the intermediate and advanced courses we teach many more rescue techniques to assist candidates in most imaginable scenarios that they may encounter. This includes rescues where the patient is below the point of safety, also known as a lifting rescue, and scenarios where the casualty is to the side of the point of safety, where you would perform a cableway or diagonal rescue.

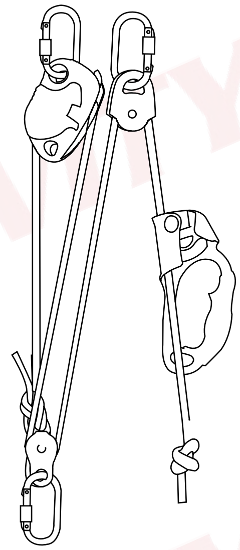
It is important for the FPP developer to understand that there are a variety of rescues available and that there is no single rescue for all scenarios. Therefore, the FPP developer will have to develop a rescue specific to the site and the access method applied.

3. Rope Rigging and Mechanical Lifting

We've covered the **people** part of this section. With one of the methods discussed above, we should be able to get our workers to the point where the work is done safely. But what about the **tools** and **equipment** the worker needs? Can we reasonably expect a person to carry a 40 kg load with them while climbing, for instance?

Certainly not. That is why we have methods in this industry for lifting and lowering loads from 9 kg to 100 kg, by means of mechanical advantage systems. This is called basic rope rigging, otherwise known as manual lifting. In basic rope rigging, we teach the candidates the different mechanical advantage systems as well as how to lift and lower with the use of a progress capture device, in the event of something unforeseen happening and the haulers letting go of the rope.

For loads exceeding 100 kg, it is fair to say that the team required to lift that type of load would need to be incredibly big. You would need to have five or six people pulling on the rope, not to mention the person on top and the banksman. Lifting such a heavy load would also be extremely dangerous if you are simply relying on manpower.



As such, mechanical lifting was born, for safely lifting heavier loads with less people while putting less strain on the people doing the lifting. With mechanical lifting, we can lift loads of up to 500 kg with ease, and it is incredibly safe to do so if you have the proper training and equipment.

However, lifting and lowering activities still subject technicians and other people onsite to certain risks. Therefore, the following mitigations must be in place:

- Lifting and lowering must be conducted in accordance with a **lifting plan**, written by a competent person.
- Lifting and lowering must be conducted by competent persons.
- Lifting and lowering must be conducted in teams of at least two (depending on the type and number of systems to be used).
- Barricading and drop zones are to be used as needed.

Lifting and lowering risks can pose fatal should it be attempted by incompetent persons. The risk assessment and training matrix must stipulate sufficient mitigations.

All these components form the scope for working at heights. Though the structures may change as we move from lattice structures to monopoles, rooftops, the side of a building, scaffolding, poles and ladders, all the way to openings in the ground, the general application of safe access can be done using one or a combination of the methods discussed. As long as you do proper planning, nothing is impossible.

With fall prevention and protection planning, we aim to do just that – we plan to use the correct access method, or combination of access methods, in order to perform the task at hand quickly, professionally and safely.

Proper planning prevents poor performance.

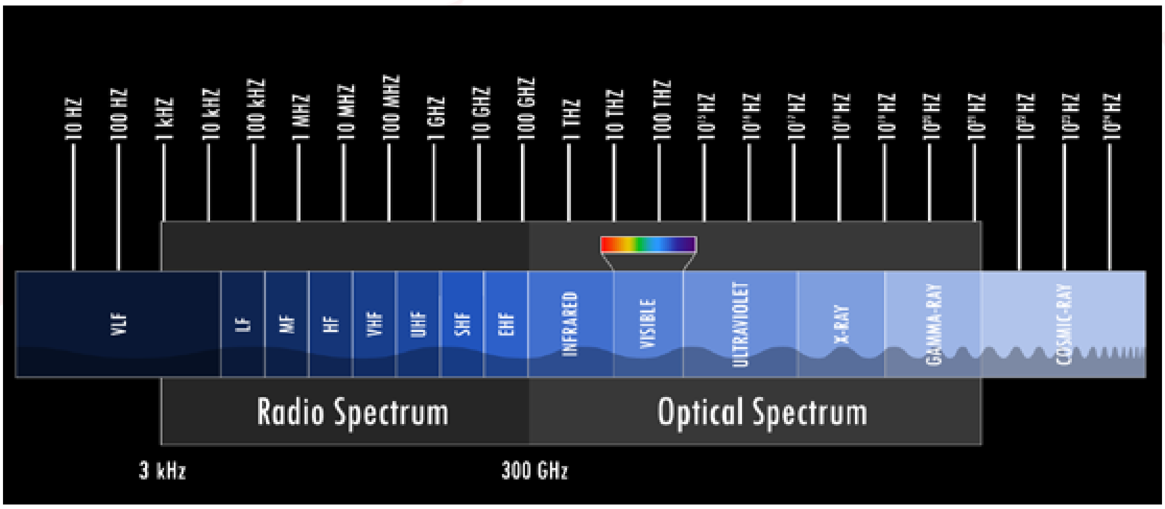
Let's look at some other factors that could impact our work.

4. Radio Frequency Awareness

Due to the increase in use of electronic devices, the amount of radio frequency (RF) radiation has been increasing significantly, resulting in most open spaces being used to erect antennas and/or dishes for the broadcasting and/or receiving of RF signals. The radiation given off by such devices poses a significant risk to workers, for example:

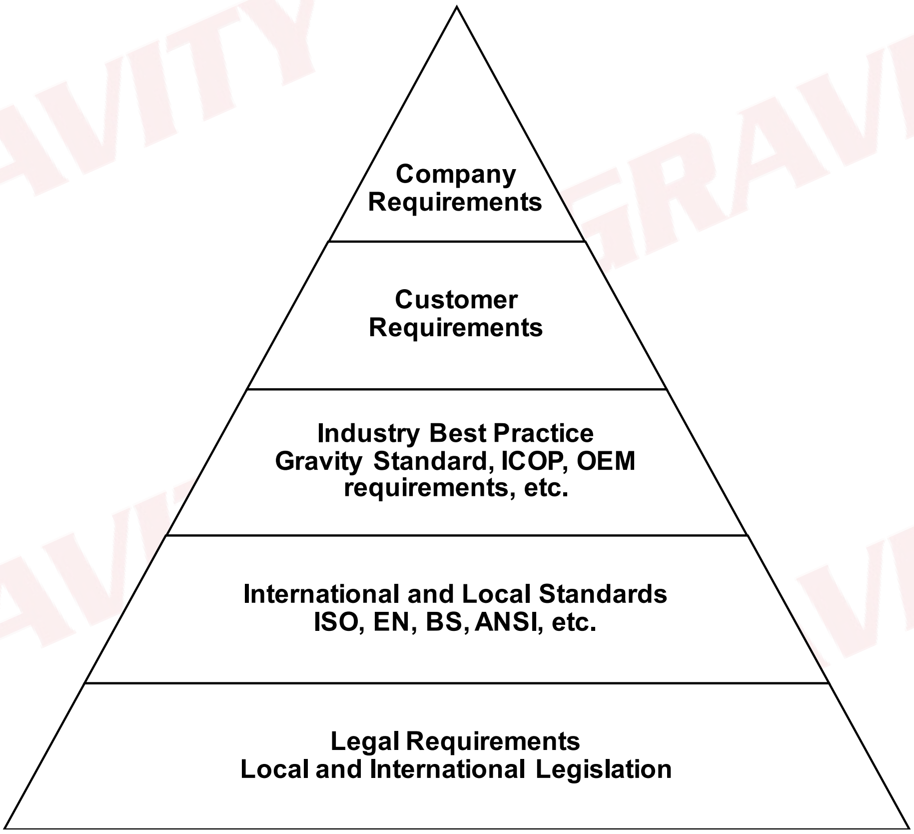
- Prolonged exposure can cause serious internal and external injury.
- Unfamiliarity with equipment and/or procedures may cause workers to suffer electrical shocks/burns.
- RF radiation can cause eye injuries.

It should be clear that RF radiation adds significant risks to a worksite, and should thus be mitigated by ensuring that all work carried out in RF areas is done so by trained and competent technicians. Ensure that this is included in the training matrix.



5. Control Hierarchy

In order to manage the access methods, we need to make use of a control hierarchy. When we are developing our FPP, it is important to look at what we need to comply with. This will instruct us on what we need to do as a minimum to use as a foundation to build from.



When writing an FPP, we always begin from the bottom of this pyramid, starting with what is legally required, then moving to what is internationally required, what the industry is doing, what our customer requires and what we as a company are committed to.

This hierarchy can also flow from the top down, though. If a customer likes something you do as a company, they can incorporate it into their standard practices. If enough of your customers choose to implement something you are doing above and beyond their requirements, it can become an industry best practice. Industry best practices are often written into international and local standards to improve the quality of the standards, and governments may choose to incorporate standards into legislation.

This means you could effectively cause a change in legislation in your country by being proactive and innovative in your work.

i. Legal Requirements

It is crucial to start off with what is legally required in your country. You cannot opt to do less than what is legally required, and as such, you should start with local legislation. This serves as a baseline for your fall protection plan and the actions that you will commit.

Let's have a look at what legislation in South Africa says in terms of working at heights.

OHS Act

- Section 8 – Duties of the Employer
- Section 14 – Duties of the Employee
- Section 24 – Duty to Report
- Section 44 – Incorporation of Standards

General Safety Regualtions

- GSR 3 – Emergency Preparedness
- GSR 6 – Work from Elevated Positions
- GSR 13 – Ladders

Construction Regulations
- CR 10(1) – FPP Requirements
- CR 10(2) – FPP Minimum Content Requirements
- CR 10(3) – Implementing an FPP
- CR 10(4) – Securing Holes and Openings
- CR 10(5) – Roofwork
- CR 7(8) – Medical Evaluation
- CR 8 – Designations and Roles
- CR 9 – Risk Management
- CR 11 – Structures
- CR 16 – Scaffolding
- CR 18 – Rope Access

Driven Machinery Regulations
- DMR 18 – Lifting Machines, Hand-Powered Lifting Devices and Lifting Tackle

ii. International and Local Standards

At Gravity, we incorporate international standards into our work because we work with customers that operate on an international scale. This enables us to assist a larger market with their work-at-height needs and helps our customers compete on a higher level. These standards include but are not limited to:
- ISO 22846-1 and ISO 22846-2 (Rope Access Standards)
- ANSI (A10.48, Z359, A490, etc.)
- EN Standards
- British Standards
- IRATA, ICOP and TACS
- OEM Specifications

iii. Industry Best Practice

Industry best practice refers to what the industry is doing as a norm and what is considered reasonable. If most of the industry is doing something for the same reason, it is reasonable to assume that it is something everyone can do.

For example: If the majority wear a specific kind of shoe because that shoe is safer, it becomes industry best practice to make use of that particular type of shoe. Such an expectation is deemed reasonable because the majority applied it willingly.

iv. Customer Requirements

When developing your FPP, it is critical to look at what your customer requires. Customers may require more than what is expected of you in the aforementioned three building blocks if it would serve to promote a safer work environment. However, customers cannot expect you to do less than what is legally required.

Even if you are legally compliant, though, a customer may still reject your plan if you are not willing to rise to their level of requirments. If they have made a commitment to maintain a higher standard of safety, they may choose not to work with your company if you are not willing to make the same commitment.

v. Company Requirements

The last building block to an effective FPP is what you as a company are comitted to. This is very important because you must follow through on what you say you will do. Do not commit to something just to make your plan stand out if you are not willing to see it through.

From a legal standpoint, if your policies and procedures include a commitment to take certain actions and an investigation reveals that you failed to fulfill that commitment, you and your directors may be held legally accountable.

6. Policies and Procedures

Policies, procedures and plans make up the framework for running a business. Typically, you would have policies relating to all aspects of your business, like driving, electrical work, incident reporting and (in this case) work at height. Do you know the difference between a policy, a procedure and a plan?

What is a policy?
A statement or promise that your business makes relating to a specific topic.

What is a procedure?
Specific steps that need to be followed; items to be used and processes to be applied.

What is a plan?
A combination of policies and procedures.

A company must provide proof of their commitment to conduct work in accordance with the relevant requirements. This is separated into two components:

1. A planning component to address what will be done before going to site.
2. An implementation component to address what will be done onsite.

To ensure all aspects are covered for both components, a policy or procedure containing the following should be written:

i. Planning

- A reason – **Why (logically)** the policy exists and **why (legally)** it must be adhered to.
- A responsible person – The person **who** will be charged with ensuring that the policy is implemented. It is recommended that the person is not named, but rather referred to by designation. This will prevent having to rewrite the plan whenever there is a change of personnel.
- Task – Put quite simply, this is **what** must be done by the responsible person. Ensure that no task contradicts any of the regulations or standards mentioned previously. Refer to the relevant regulations where applicable.
- **How** will refer to what guidelines will be followed to apply the actions stated in the **what** section. There are usually requirements in legislation and standards for how certain actions need to be completed or what must be contained in a procedure.
- Time – It must be stipulated **when** a given task will be done by a responsible person. Typically, most policies must be implemented "before work starts". Again, be careful not to be too specific with these stipulations, as this may lead to liability issues in the case of an incident or accident.

ii. Implementation

- **What** will be done onsite?
- **Who** will be responsible for doing it?
- **When** will they do it?
- **How** will they <u>prove</u> that they have done it?

An example of a risk assessment policy:

To identify, manage and mitigate all fall risks to ensure and maintain a safe work environment, as per CR 10(2)(a) and Sect. 8, the FPP developer will develop a fall risk assessment in accordance with CR 9(1), before the project starts.

The site supervisor will conduct and communicate an onsite risk assessment to all persons working from a fall risk position, before work starts onsite, and proof thereof shall be kept in the HSE file

In the example above:
For **planning**:

- The **why (logical)** is to create and maintain a safe work environment.
- The **why (legal)** is as per CR 10(2)(a).
- The **who** is the competent FPP developer.
- The **what** is the fall risk assessment.
- The **how** is in accordance with CR 9(1).
- The **when** is before work starts.

For **implementation**:

- The **who** is the site supervisor.
- The **what** is conduct and communicate all fall risks onsite.
- The **when** is before work starts.
- The **proof** will be the physical risk assessment in the HSE file onsite.

iii. Common Phrases, and Understanding Your Intended Reader

While reading the example, you may have noticed the phrases "as per" and "in accordance with". These two phrases are commonly used and mixed up in policies and procedures.

- "As per" refers to where it is stated that an action must be taken. Think of the guy that always says "as per the boss".
- "In accordance with" refers to the guideline you will use to measure compliance. It translates to "it will be done the same way as".

While writing policies and procedures, you will also work with legislation quite a lot. Part of this entails understanding what is written in the law and what it was meant to do. This is where the phrase "the letter of the law" comes from and refers to literally doing what is stated in the law. The same would apply to your policies and procedures. You should write them in such a way that literally doing what is described will result in the safest working environment possible. When writing policies and procedures, you must be specific in the "what needs to be done and how". Therefore, you must be able to answer certain questions when you are done reading that policy or procedure. Be sure not to write it in such a way that it can be interpreted incorrectly or be subjected to malicious compliance.

On the other hand, there is "the intent of the law". Where the law can be misinterpreted or mean more than one thing, it is important to look at what the aim of the law was. The aim can take precedent above what is written as it was initially written to achieve a specific outcome. That is why intent is so important. Also, remember who you are writing this policy or procedure for. If you try to write it in a legislative style, it can be misunderstood or misinterpreted by your audience. Keep it simple and be clear on what the desired outcome is.

iv. Standard Operating Procedures

The SOPs specify the exact steps that need to be taken to perform a given task. It is recommended that the personnel involved in performing the tasks are included in the development process of the SOPs.

SOPs are task- and site-specific. Sometimes it may be possible to write one SOP that can be applied to various sites, e.g. using fall arrest equipment will always be done as per the training matrix. However, this might not always be the case. Thus, ensure that all SOPs are applicable to the relevant site.

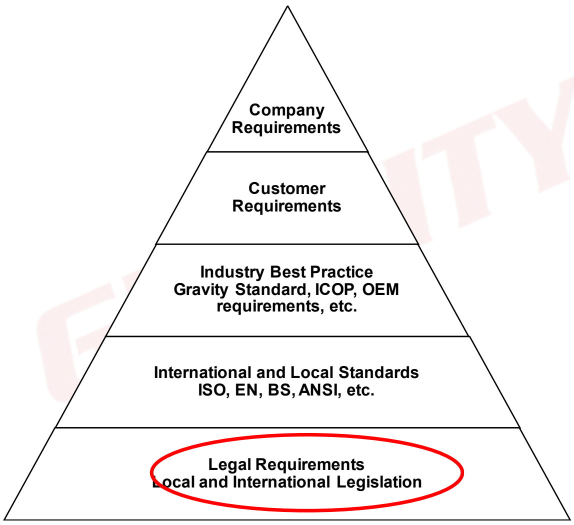
SOPs must include the following as a minimum:

- Scope and purpose – Describe the reason for the existence of the SOP and the affected personnel/departments.
- Equipment/Supplies/Materials – Indicate what external items are required to complete a given task.
- Methodology – List all the steps with the necessary detail to complete a given task.
- "What-ifs" and interferences – State what can go wrong and what steps need to be taken should something outside of the SOP occur.
- Review process – SOPs need to be reviewed in accordance with the company's normal review schedule, as well as after an incident or accident has occurred. It is also recommended that the persons involved in carrying out the SOP are included in the review process.

7. Fall Protection Plan Requirements

Let's start applying what we've learned by planning a policy for our FPP. As mentioned, we need to look at the control hierarchy and build according to what is required.

Legal Requirements



Construction Regulation 10(1)

A contractor must:

a. designate a competent person to be responsible for the preparation of an FPP,

b. ensure that the FPP contemplated in paragraph (a) is implemented and amended as required, and

c. take steps to ensure continued adherence to the FPP.

Construction Regulation 10(2)

An FPP as contemplated in sub-regulation (1), must include:

a. a risk assessment of all work carried out from a fall risk position and the procedures and methods used to address all the risks identified per location;

b. the processes for the evaluation of the employees' medical fitness necessary to work at a fall risk position, and the records thereof;

c. a programme for the training of employees working from a fall risk position, and the records thereof;

d. the procedure addressing the inspection, testing and maintenance of all fall protection equipment; and

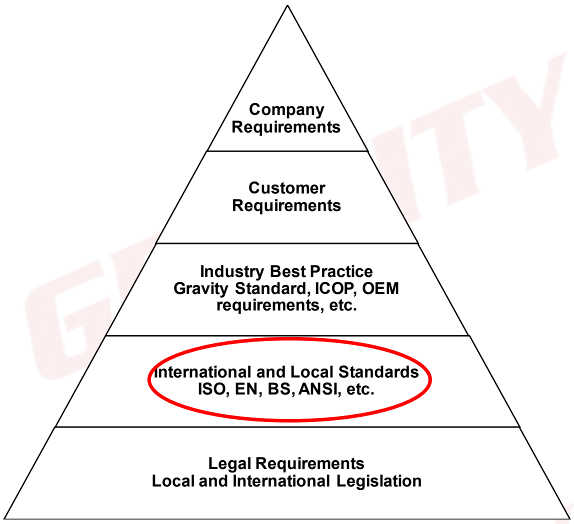
e. a rescue plan detailing the necessary procedure, personnel and suitable equipment required to effect a rescue of a person in the event of a fall incident – this ensures that the rescue procedure can be implemented immediately following the incident.

Construction Regulation 10(3)

A contractor must ensure that a construction manager appointed under regulation 8(1) is in possession of the most recently updated version of the FPP.

Legal requirements have provided us with all the information we require for our FPP policy. However, it is still a good idea to look at the rest of these building blocks to ensure we build the best possible FPP.

International and Local Standards

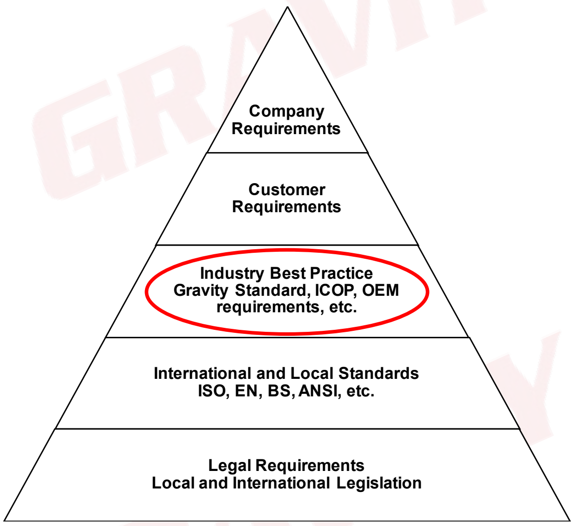


Our next building block is international standards. International standards are very important, especially when working with customers that operate in more than one country.

ANSI A10.48 Sect. 6.3 makes specific reference to the requirement of an FPP and the content in said plan, which includes:

- the type of structure,
- emergency contact information,
- known hazards,
- competent and authorised climbers,
- access methods,
- climbing obstructions,
- transition methods,
- equipment, and
- permanent and/or temporary anchors.

Industry Best Practice



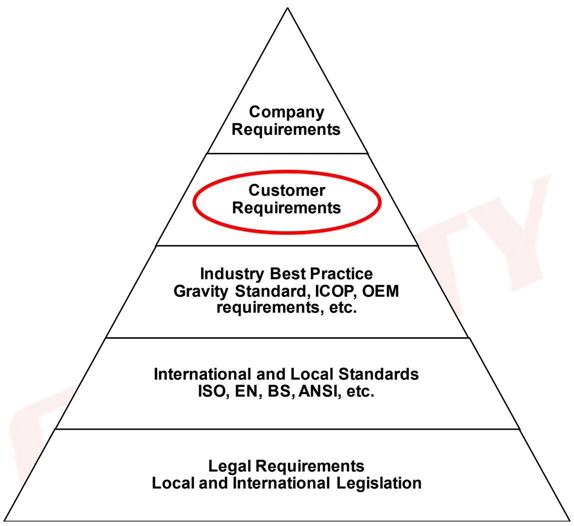
Following international standards, we look at what the rest of the industry is doing.

What the rest of the industry is doing is generally a great guideline for what you as a business should do, because in the event of an incident, what the industry is doing is usually referred to as common or reasonable practice.

In the work-at-height industry, it has become a common requirement to plan work at height in advance, and this plan must include as a minimum:

- a fall risk assessment,
- a medical evaluation procedure,
- a training programme and training matrix,
- an equipment management procedure and
- a site-specific rescue plan.

Customer Requirements



Then, we also consider customer requirements.

It is important to consider that, above and beyond legal, international and industry requirements, a customer may also require that you have certain elements in place to prevent persons and/or objects from falling.

These requirements must be considered when developing an FPP. A customer may expect more of your company than what is required by the legal, international and industry standards, but the company cannot expect you to do less.

8. Fall Protection Plan Structure

Now that we know the basic requirements of an FPP, how do we proceed? With the information we have, we can develop our FPP policy. Here we start assigning roles and responsibilities to people.

Our next logical step would be to do a risk assessment to determine what hazards workers will be exposed to and how things could go wrong. But do we know who will conduct the risk assessment, how it will be done, why it's necessary and when it will take place? If the answer is no, we need to create a risk management plan first.

When we have identified the hazards that workers will be exposed to, we can continue to write more procedures to address these hazards. That means our FPP will include the following:

- FPP policy
- Risk management plan
- Risk assessment and DSTI

Next, we will address any additional legal requirements, followed by procedures for the items identified in our risk assessment:

- Medical evaluation procedure
- Training programme and training matrix
- Equipment management procedure
- Rescue plan
- Monitor and review plan
- Additional Procedures

Activity 1

Write a policy summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; w**hen** will this be implemented; and how will they **prove** it has been implemented?

C. RISK MANAGEMENT PLAN

The work-at-height risk assessment forms the basis of the FPP, and as such, it is of utmost importance that it is conducted in such a way that it allows a company to establish and maintain a safe work environment with as little effort as possible.

Thus, when conducting a work-at-height risk assessment, place particular emphasis on the following:

- The work-at-height risk assessment must be completed by a designated competent person.
- Ensure that as many hazards and risks are identified as possible in order to be able to prevent any surprises when a site is established.
- Risk mitigations must be practical and reasonable as far as possible.
- Risk mitigations must not introduce any new risks onto a site. If a risk mitigation does introduce a new risk, e.g. using fall arrest, the new risk must be mitigated similarly to all other risks.
- Ensure that the process of assigning risk ratings is documented (in a risk matrix) and understood by all persons involved in the risk assessment process.
- Beware of writing impossible-to-implement risk mitigations.
- Be cautious of writing mitigations in a way that could make the company liable for incidents. For example, stating that workers will stop for lunch at 12:00 could expose the company if a worker falls at 12:05. It's better to phrase the mitigation more flexibly. Consider using more adaptable terms such as "as needed" instead of a specific time like "12:00".
- An issue-based risk assessment must be conducted per site and task to ensure the relevant risks onsite are identified and mitigated. This is to be completed by the FPP developer before work starts.

It is the duty of the FPP developer to ensure that the following is identified:

1. The Prevalent Hazards Onsite

- Falling from height.
- Workers dropping tools or equipment from height.
- Tools or equipment falling onto people below.
- Dangerous areas where people are not allowed (open edges, holes, fragile surfaces, etc.).
- Conditions onsite that may have an impact on work-at-height activities (inclement weather, radio frequency, overhead powerlines, etc.).

2. The Risk Controls Established for the Above-Mentioned Risks

What are the needed steps to be taken to reduce the risks to an acceptable level? The implementer (site supervisor) must identify the risks and ensure that the relevant baseline risk assessment controls are implemented to ensure a safe work environment. This is indicated on the pre-task risk assessment.

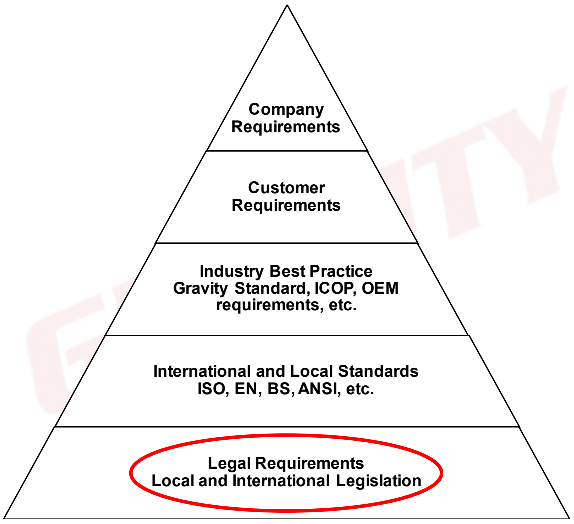
3. The Risk Owners

The persons responsible for ensuring that the risk controls/mitigations are implemented and continuously adhered to. The risk owners may vary from the employer to the employee, depending on the risk and the controls, but in most cases, the **FPP developer** is responsible for the **initial development** of procedures and the **site supervisor** is responsible for the **onsite implementation** of those procedures.

All risk assessment training and induction must be recorded, and the records must be kept.

All the above is accomplished with FPP induction training and DSTI or pre-task risk assessments. It is recommended that DSTI be monitored at least weekly by a competent person.

Legal Requirements



Construction Regulation 10(2)(a)
A risk assessment of all work carried out from a fall risk position, and the procedures and methods used to address all the risks identified per location.

Construction Regulation 9(1)
A contractor must, before the commencement of any construction work and during such construction work, have risk assessments performed by a competent person appointed in writing, which risk assessments form part of the health and safety plan to be applied on the site, and must include-

a. the identification of the risks and hazards to which persons may be exposed to;
b. an analysis and evaluation of the risks and hazards identified, based on a documented method;
c. a documented plan and applicable safe work procedures to mitigate, reduce or control the risks and hazards that have been identified;
d. a monitoring plan; and
e. a review plan.

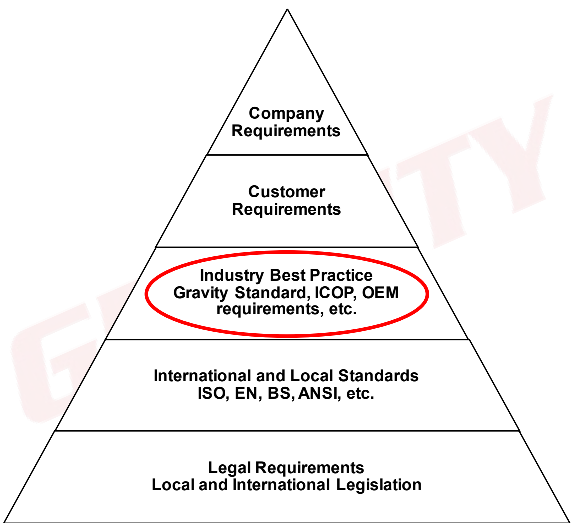
International and Local Standards

Our next building block is international standards. International standards are very important, especially when working with customers that operate in more than one country.

The method of performing a risk assessment is prominent in almost all international standards and refers to the principles we cover in our training:

- Identify
- Analyse and evaluate
- Mitigate
- Monitor and Review

Industry Best Practice



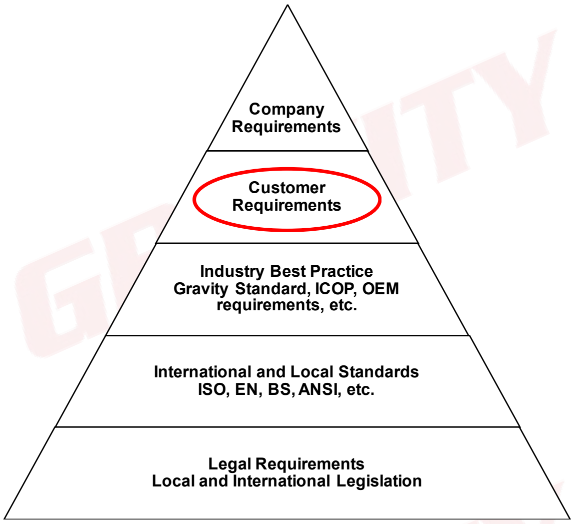
Following international standards, we look at what the rest of the industry is doing.

What the rest of the industry is doing is generally a great guideline for what you as a business should do, because in the event of an incident, what the industry is doing is usually referred to as common practice or reasonable practice.

The current industry best practice relating to risk assessments is a project or baseline risk assessment, performed by a competent person before work starts.

Further to this, it is also required that an onsite risk assessment is conducted by the site supervisor (who must be a competent risk assessor) and communicated to all persons onsite before work starts.

Customer Requirements



Then, we look at customer requirements.

A customer may require more frequent risk assessments and even a different methodology. For example, a customer may require the Injury-, Production-, Environmental-, Cost Impact Factor (IPEC) methodology as it also caters for a cost impact.

You may be required to complete a risk assessment on the cuctomer's structure, despite having your own risk assessment in place.

Activity 2

Write a risk management plan summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; **when** will this be implemented; and how will they **prove** it has been implemented?

D. RISK ASSESSMENT

As discussed in our risk management plan, there are certain criteria that need to be met for our risk assessment to be effective and meet the local, international, industry and customer requirements. The criteria are as follows:

- Identify the source of danger.
- Identify how that source of danger can harm someone or cause property damage.
- Analyse and evaluate the hazard and the harm it can cause.
- Mitigate the source of exposure to reduce the likelihood of occurrence as much as you can.
- Monitor the implementation of controls and review the effectiveness.

Combined, these form an effective risk assessment. All this information has been placed into an easy-to-read structure below.

Identify the source of danger and the possible harm that it can cause.		Analyse and evaluate the hazard and the harm that it can cause.			Mitigate the source of exposure to reduce the likelihood of occurrence as much as you can.	Monitor the implementation of controls and review the effectiveness		
Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk

1. Identifying a Source of Danger

Psychology research has uncovered an interesting phenomenon that relates to the "I-didn't-see-you" excuse. It seems that humans are subjected to so much incoming information that one part of the brain serves as a sort of subconscious mental "spam filter". Thus, the conscious part of the brain only receives information that corresponds to what the person is concerned about.

If a driver is concerned with buying a new car, the spam filter causes that anything related to cars, including automobile dealerships and billboards with car ads, will pass through to the driver. The flip side is that if the driver is not concerned about something (say motorcycles), then the spam filter may delete those images. So it really can be true that a driver looks right at you and sees you, but your image gets filtered out and never reaches the conscious part of the driver's brain.

That would explain how a driver might not comprehend a motorcyclist wearing a high-visibility jacket and flashing his bike's headlights – or, for that matter, a freight train with flashing headlights and air horns blaring. Onsite you might think of things like climbing but ignore factors like veld fires, a strike in a densely populated area, etc.

Thus, it is important that workers condition themselves to be attentive to any and all factors that may influence their safety or the safety of those around them. Workers should enter a worksite with the mentality that safety is the highest priority – safety must, therefore, be at the forefront of the conscious mind.

Taking all of this into account, there is a certain way that we need to phrase a hazard/source of exposure. Then, we want to consider when this hazard/source of exposure will be an issue. For example:

Often risk assessors indicate "medical fitness" as a hazard. However, medical fitness is not a hazard. In fact, it is what we are trying to achieve. Not being medically fit is the hazard. But does it matter if the receptionist is not medically fit to work at height? Her job will never require her to climb. So, when would medical fitness be a concern? Let's consult our structure.

Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk
Not being medically fit while working from a fall risk position.								

It is important to explain how the source of exposure can cause harm and what that harm is. Often we will see that risk assessors only write "serious injury or death". Serious injury or death is indeed the harm our source of exposure (i.e. falling) can cause, but how did we get from not being medically fit to dying? Something happens as the result of not being medically fit which causes a fall, and that fall is how we sustain injuries, die or cause damage to property. After all, this course is called **Fall** Protection Plan Developer.

Let's see how this would look in our risk assessment.

Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk
Not being medically fit while working from a fall risk position.	An unknown medical condition could result in a person losing consciousness and falling. Such a fall can lead to serious injury, death and/or property damage.							

2. Analysing and Evaluating

The next step is to "score" our risk. We do this by determining how likely it is that the hazard will arise and cause harm, and then we look at how severe the harm will be.

To do this, we need a risk matrix. A risk matrix helps us to determine the level of risk by factoring in the likelihood of occurrence and severity of outcome.

Let's look at our example.

Risk Matrix			**Consequences or Severity**				
			1	**2**	**3**	**4**	**5**
			Negligible	**Minor**	**Moderate**	**Major**	**Catastrophic**
			No Injuries	First Aid and Short Term	External, Medical or Loss Time Injury – Medium Term	Extensive or Life-Changing Injuries	Death(s) and Major Injuries
			Zero Damage	Slight Damage	Localised Damage	Major Damage	Extensive to Critical Damage
Likelihood of Occurrence	**1**	**Rare**	**LOW 1**	**LOW 2**	**LOW 3**	**LOW 4**	**MEDIUM 5***
	2	**Unlikely**	**LOW 2**	**LOW 4**	**MEDIUM 6**	**MEDIUM 8**	**MEDIUM 10**
	3	**Possible (50/50)**	**LOW 3**	**MEDIUM 6**	**MEDIUM 9**	**MEDIUM 12**	**HIGH 15**
	4	**Likely**	**LOW 4**	**MEDIUM 8**	**MEDIUM 12**	**HIGH 16**	**HIGH 20**
	5	**Almost Certain**	**MEDIUM 5***	**MEDIUM 10**	**HIGH 15**	**HIGH 20**	**HIGH 25**

The pre-mitigation risk rating is based upon the risk matrix and refers to the state of the risk **before any control measures have been implemented**. The rating is calculated as follows: Firstly, the likelihood is considered on a scale from 1 (Rare) to 5 (Almost Certain). Then, the severity is considered on a scale from 1 (Negligible) to 5 (Catastrophic). For example:

Risk Matrix			**Consequences or Severity**				
			1	**2**	**3**	**4**	**5**
			Negligible	**Minor**	**Moderate**	**Major**	**Catastrophic**
Likelihood of Occurrence	**1**	**Rare**	**LOW 1**	**LOW 2**	**LOW 3**	**LOW 4**	**MEDIUM 5***
	2	**Unlikely**	**LOW 2**	**LOW 4**	**MEDIUM 6**	**MEDIUM 8**	**MEDIUM 10**
	3	**Possible (50/50)**	**LOW 3**	**MEDIUM 6**	**MEDIUM 9**	**MEDIUM 12**	**HIGH 15**
	4	**Likely**	**LOW 4**	**MEDIUM 8**	**MEDIUM 12**	**HIGH 16**	**HIGH 20**
	5	**Almost Certain**	**MEDIUM 5***	**MEDIUM 10**	**HIGH 15**	**HIGH 20**	**HIGH 25**

5 – Almost Certain – There is not only a chance of an occurrence but a chance of frequent occurence (more than once).

5 – Catastrophic – A high severity which indicates that, should the risk remain uncontrolled, a fatality will occur. It is also important to note that because we indicated in our possible harm that the concequence could be fatal, we must assign the appropriate severity rating.

Risk Rating – If we use the risk matrix and combine these two factors –the likelihood that a person will be injured or harmed and the fact that the injury or harm will be serious – we arrive at the final pre-mitigation risk rating of **HIGH 25**, which indicates that no work may commence or continue without additional controls being implemented.

Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk
Not being medically fit while working from a fall risk position.	An unknown medical condition could result in a person losing consciousness and falling. Such a fall can lead to serious injury, death and/or property damage.	5	5	25 H				

3. Reducing the Risk

We now know that our risk is very high. It is our duty as FPP developers to put mitigations in place to either reduce the likelihood, the severity or both.

We do this by assigning someone to a responsibility that must be completed by a specific time. Who will do what, and when will it be done? These mitigations can be proactive or reactive depending on what it will reduce.

- **Proactive mitigations** typically reduce the likelihood of an occurence by aiming to prevent an encounter with the hazard. An example of proactive mitigations would be to "stop work procedures", because this removes the workers from the source of exposure.
- **Reactive mitigations** typically reduce the severity of an occurence by aiming to protect in the event of an encounter with the hazard. An example of reactive mitigations would be the use of fall arrest equipment as it would not prevent workers from falling, but it would make the outcome of the fall less severe should a fall occur.

4. Mitigation Hierarchy

When mitigating hazards, we use the mitigation hierarchy to reduce either the likelihood of occurrence or the severity.

- The first step is to avoid the risk. For example, stay away from an area where there is a risk of falling.
- If we can't avoid the risk, we try to prevent exposure to the risk. For example, installing barricading to isolate the risk area.
- If isolating the risk is not an option, we try and change the process to see if we can eliminate work at height as a component. However, that is not always possible.
- If we can't eliminate work at height, we try to use measures to prevent falling (i.e. work restraint).
- If work restraint is not possible or won't solve the problem (for example, when working on a tower), we need to explore alternative access methods that allow us to perform the job safely, such as fall arrest systems.
- If no other access methods will allow us to work safely, we stop the work completely.

Part of the success of the mitigation hierarchy is the "just culture" principle. Just culture promotes learning from unsafe acts wherever they are observed, improving safety and awareness through better recognition of potential hazards.

Competent persons and employees have accepted the fact that no fall risk environment can be made completely risk-free. Therefore, to implement an FPP, all employees and the company need to ensure that they have completed the risk controls correctly.

Every individual must accept the following responsibilities to reduce fall risks to a reasonable and practicable level of safety:

- The duty to avoid causing unjustified risk or harm during the task.
- The duty to produce an outcome.
- The duty to follow a procedural rule (i.e. an SOP).

By applying the just culture principle in our organisations, we can ensure effective controls, as this principle relies on people fulfilling their commitments.

The last component to consider is the "no-lone-work" rule. This rule is based on the principle that there must always be at least two workers onsite, each capable of rescuing the other. Thus, in terms of mitigations per hazard, we will have the following four types:

- **Planning Mitigation** – What will be done before we go to site?
- **Implementation Mitigation** – What will be done onsite?
- **Just Culture Mitigation** – What is each employee's responsibility in terms of preventing exposure to the hazard?
- **Team Composition Mitigation** – Will we ensure that there are always at least two employees onsite who are capable of performing a rescue?

Let's apply what we've learned to our risk assessment.

Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk
Not being medically fit while working from a fall risk position.	An unknown medical condition could result in a person losing consciousness and falling. Such a fall can lead to serious injury, death and/or property damage.	5	5	25 H	• FPP developer will develop a medical evaluation procedure before work starts. • Site supervisor to ensure workers have valid medical certificates before work starts. • Employees to report any changes in condition to the supervisor immediately. • Buddy system to always be applied.			

5. Monitor and Review

The final step is to monitor the effectiveness of our controls (PTOs or planned job observations) and review them if we see that the likelihood or severity of the risks can be reduced. In many cases, the likelihood is the determining factor. In our example, if a person falls as a result of medical unfitness, they die. But what happens if everything is in order and they still fall?

It is important to remember that if the event still occurs, it is very likely that the outcome will still be the same. If a person has a valid medical certificate and still falls as the result of a medical condition, it is likely that they will still die.

Once we have a risk rating of **LOW**, we are satisfied that we can now work, provided that the controls are implemented. If we cannot reach a rating of **LOW**, we try and revisit our controls to see what else we can do.

Let's look at our example.

Risk Matrix			**Consequences or Severity**				
			1	**2**	**3**	**4**	**5**
			Negligible	**Minor**	**Moderate**	**Major**	**Catastrophic**
Likelihood of Occurrence	**1**	**Rare**	**LOW 1**	**LOW 2**	**LOW 3**	**LOW 4**	**MEDIUM 5***
	2	**Unlikely**	**LOW 2**	**LOW 4**	**MEDIUM 6**	**MEDIUM 8**	**MEDIUM 10**
	3	**Possible (50/50)**	**LOW 3**	**MEDIUM 6**	**MEDIUM 9**	**MEDIUM 12**	**HIGH 15**
	4	**Likely**	**LOW 4**	**MEDIUM 8**	**MEDIUM 12**	**HIGH 16**	**HIGH 20**
	5	**Almost Certain**	**MEDIUM 5***	**MEDIUM 10**	**HIGH 15**	**HIGH 20**	**HIGH 25**

NOTE:
HIGH RISK (15–25) – No work if risk is **HIGH**.

MEDIUM RISK (6–12) – Work may only continue with controls as approved by a subject matter expert.
*Note: **MEDIUM (5)** risk work is acceptable if adequate proof is available that controls are correctly implemented before work starts.

LOW RISK (1–4) – Work may start, but the area, task and tool use must be monitored by all, and any changes must be reported immediately.

1 - Rare – There is almost no chance of an occurrence if our mitigations are adhered to.

5 - Catastrophic – A high severity which indicates that, should the risk remain uncontrolled, a fatality will occur. It is also important to note that because we indicated under “possible harm” that the concequence could be fatal, we must assign the appropriate severity rating.

Risk Rating – If we use the risk matrix and combine these two factors – the likelihood that a person will be injured or harmed and the fact that the injury or harm will be serious – we arrive at the final post-mitigation risk rating of **MEDIUM 5**, which indicates that work may commence or continue provided that there is adequate proof that the controls were implemented before work starts.

Hazard (Source of danger; When is it an issue?)	Harm (How does the hazard cause harm and what is the harm?)	Likelihood	Severity	Risk	Mitigation (Who will do what, when?)	Likelihood	Severity	Risk
Not being medically fit while working from a fall risk position.	An unknown medical condition could result in a person losing consciousness and falling. Such a fall can lead to serious injury, death and/or property damage.	5	5	25 H	• FPP developer will develop a medical evaluation procedure before work starts. • Site supervisor to ensure workers have valid medical certificates before work starts. • Employees to report any changes in condition to the supervisor immediately. • Buddy system to always be applied.	1	5	5 M

Activity 3.1

Indicate the four hazards that may be present if we do not have the ADMINISTRATIVE controls in place. List these hazards below and then complete a risk assessment for these hazards.

Complete the practice activities as preparation for your assignment!

Remember:
The minimum requirements for an FPP is a medical evaluation procedure, training programme, equipment management procedure and rescue plan.

Tip:
The primary hazard is not the lack of procedure in itself, but rather what happens **because** there is no procedure (i.e. workers don’t know what to do in case of a fall). For example, if a fall occurs and there is no rescue plan, a patient that has fallen could hang for too long while waiting to be rescued, which would have catastrophic consequences.

ADMINISTRATIVE HAZARDS

Hazard	Possible Harm and/or Property Damage	Likelihood	Severity	Risk	Controls and Preventative Measures (Who will do what, when?)	Likelihood	Severity	Risk

ADMINISTRATIVE HAZARDS								
Hazard	Possible Harm and/or Property Damage	Likelihood	Severity	Risk	Controls and Preventative Measures (Who will do what, when?)	Likelihood	Severity	Risk

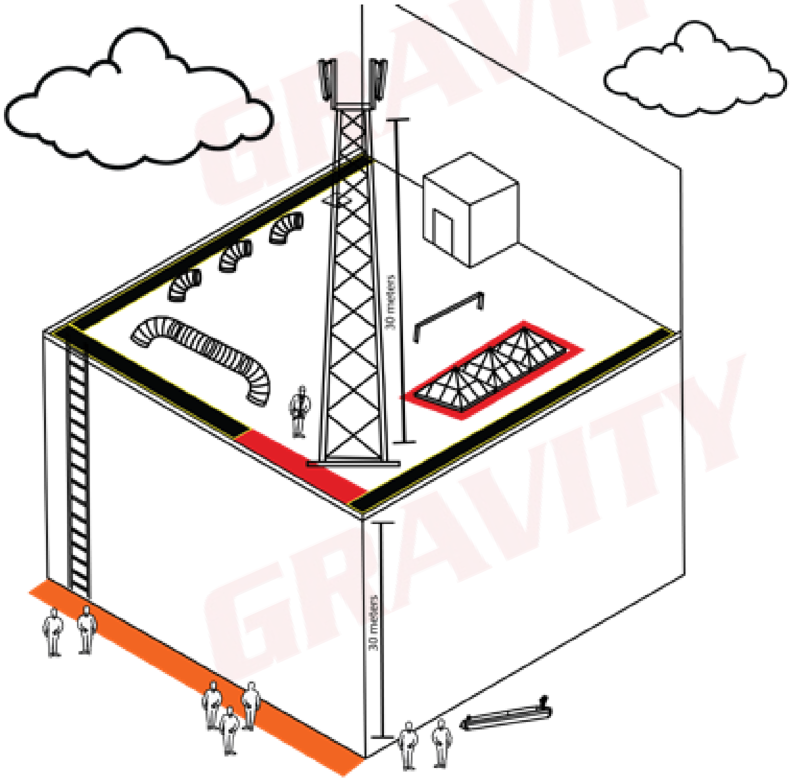
Activity 3.2

Complete a project risk assessment for ONE of the model sites. You must identify five ACTUAL hazards. List the hazards next to each site. For each hazard there must be four mitigations (planning, implementation, just culture and team composition).

Site 1

FALL ARREST TASK: Lift and install an 80 kg antenna on a 30 m telecoms structure on a rooftop. You will have four workers onsite.

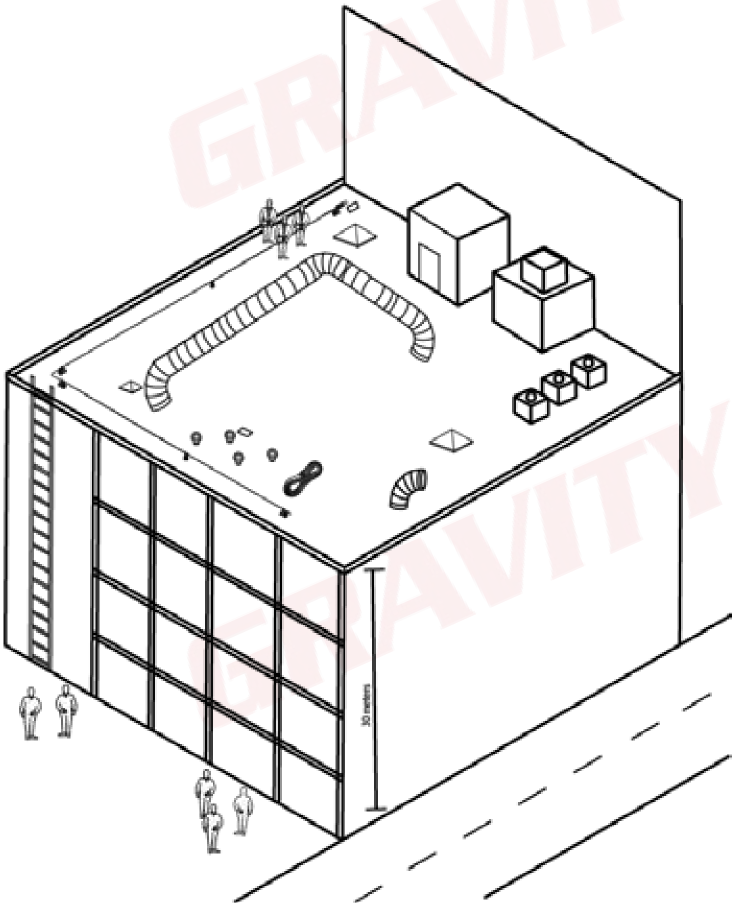
Actual Hazards



Site 2

ROPE ACCESS TASK: Washing windows that are 30 m high. You will have three workers onsite.

Actual Hazards



ACTUAL HAZARDS								
Hazard	Possible Harm and/or Property Damage	Likelihood	Severity	Risk	Controls and Preventative Measures (Who will do what, when?)	Likelihood	Severity	Risk

ACTUAL HAZARDS

Hazard	Possible Harm and/or Property Damage	Likelihood	Severity	Risk	Controls and Preventative Measures (Who will do what, when?)	Likelihood	Severity	Risk

ACTUAL HAZARDS								
Hazard	Possible Harm and/or Property Damage	Likelihood	Severity	Risk	Controls and Preventative Measures (Who will do what, when?)	Likelihood	Severity	Risk

E. DAILY SAFE TASK INSTRUCTION

Now that we have a project risk assessment, we need to make sure that the team onsite conducts a daily risk assessment and takes the appropriate steps to mitigate the risks identified – a DSTI serves to do exactly that. This will be the implementation part of our mitigations listed in our risk assessent. The DSTI should contain the following:

- Basic site information and scope of work.
- Activity list/methodology.
- Emergency information.
- Rescue information.
- Hazards identified in the project risk assessment – hazards must be communicated and mitigation must be ensured (leave space for additional hazards that may not have been identified).
- Register to sign as proof of induction.

Activity 4

Complete a DSTI for the site selected in Activity 3.2.

Complete the practice activities as preparation for your assignment!

Remember:
Some of the hazards identified in the project risk assessment does not reflect on the DSTI; so, you might need to add them.

DSTI (complete this for the site selected in Activity 3.2):

Site area		Date	
Site task area			
Permit number(s)			
Work-at-height risk assessment no.			
Supervisor name		Contact no.	
SCOPE OF WORK			
Describe task to be performed:			
List the relevant steps of the task			
1.		2.	
3.		4.	
5.		6.	
7.		8.	
9.		10.	
First aider/rescuer	1:	2:	3:
EMS no.		Police no.	
Fire/Rescue team no.		Company contact details	
How will EMS access site?			
SITE-SPECIFIC RESCUE PLAN			
Rescue method:	☐ First aid kit	☐ Rope grab(s)	☐ Descending device(s)
	☐ Ratchet rescue kit	☐ Rescue sling(s)	☐ RA lift-and-lower kit
	☐ Stretcher and basket	☐ Plug-in for tensioned lines	☐ RA lowering kit
	☐ Cableway rescue kit		☐ Rig-for-rescue
Note: Site supervisor and/or competent rescuer to discuss the rescue procedure onsite as per the task requirements.			

NO.	TASK/HAZARD	MITIGATION	TICK
1.	☐ Task-specific risks/hazards	Refer to HSE plan. Site supervisor to ensure that all risks/hazards onsite are managed reasonably and practicably, including HSE concerns.	
2.	☐ Medically fit to perform task	The site supervisor to ensure that all workers are in possession of a valid COF before work starts.	
3.	☐ Rope rigging	Site supervisor to ensure that the correct SOP/method statements are followed, as per training. OEM and industry practises to be followed. Constant competent supervision is required. A lifting plan must be completed and implemented.	
4.	Work Platform ☐ Rooftop ☐ Fragile surfaces ☐ Open edges ☐ Open holes ☐ Hand railings, parapet wall or crow's nest ☐ Sloping edges ☐ Container ☐ Adequate safe anchors/fall arrest systems Other:	Site supervisor must ensure that the work platform is safe for use and fit for purpose. Appropriate fall arrest/work restraint systems to be used where required, as per training and OEM requirements.	
5.	Access Method(s) ☐ Lattice towers ☐ Portable ladders ☐ Fixed or caged ladders ☐ Scaffolding ☐ MEWPs ☐ Man cage/cradle ☐ Monopoles ☐ Rope access Other:	Site supervisor to ensure that all workers onsite have been inducted on the baseline risk assessment and are aware of the correct mitigations to ensure that a safe work area is maintained. All access methods to be used as per the latest version of the FPP and relevant SOPs. Project manager to ensure competent supervision is appointed throughout the task, as per relevant regulations and chosen access methods.	
6.	PPE Inspection ☐ PPE usage (competence and safe use) ☐ Pre-use inspection ☐ Long-term inspection	An appointed competent person must inspect all PPE as per OEM and industry best practises. The user must ensure that they inspect their PPE before use via the buddy system.	
7.	☐ Equipment inspection	The user must ensure that they inspect their equipment before use via the buddy system.	
8.	☐ Environmental	Site supervisor to ensure that no work is conducted in adverse weather conditions or any other unsafe environmental conditions, and employees to report any concerns to the site supervisor.	
9.	☐ Dropping tools/items	Site supervisor to ensure that the correct barricading is in place and all tools/items are tied off. Helmets must be worn by all persons in a drop zone.	
10.	☐ Unprotected edges	Site supervisor to communicate unprotected edges to all workers before work starts and to supervise unprotected edges at all times. Site supervisor to ensure that proper barricading is in place where possible. Workers can make use of work restraint or fall arrest.	

11.	Sharp/Unsafe edges	Site supervisor to ensure that no PPE or other equipment is used near/over sharp edges/hazardous areas. If PPE or other equipment is close to sharp edges or any other hazardous area, edge protection must be used at all times.	
12.	Access/Exclusion zones	☐ Working edge hazard zone ☐ Top exclusion zone ☐ Exclusion zone at intermediate level ☐ Bottom exclusion zone ☐ Rigging anchor exclusion zone ☐ Patrol officer required Name and number:	
13.	Uninformed workers	Site supervisor to ensure that all workers onsite have been inducted on the baseline risk assessment and are aware of the correct mitigations to ensure that a safe work area is maintained.	
14.	Suspension trauma	Site supervisor and rescuer to ensure that a site-specific rescue plan is completed and any rescue is performed as quickly and efficiently as possible.	
15.	Other:		
16.	Other:		
17.	Other:		
18.	Other:		
19.	Other:		
20.	Other:		

DSTI ONSITE

Confirmation of risk assessment during each toolbox talk:

I hereby confirm that I have completed the required induction training as per the training matrix.
I commit to ensure all work done on a worksite is done in a safe environment, and to follow the required safe-use procedures.
I commit to immediately notify my employer and/or site supervisor if any task cannot be performed safely or if I become aware of any unsafe condition or action.
I commit to ensure that the risks will be safely and correctly mitigated before work continues onsite.
I commit to protect myself and those around me.
I confirm that I will inspect all PPE and any other equipment before use via the buddy system.

Name and Surname	Designation	Signature	Date
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

F. MEDICAL EVALUATION PROCEDURE

Part of the fall risk assessment must address the risk of falling from height due to medical unfitness. There must be a process for determining whether a person is medically fit to perform the specified work. Proof of medical fitness, based on the person's job specifications (driver, welder, work-at-height technician, etc.), must be provided. This usually comes in the form of a COF. It might be required that workers are subjected to more frequent medical evaluations due to confined space work, toxic gasses or hazardous chemicals.

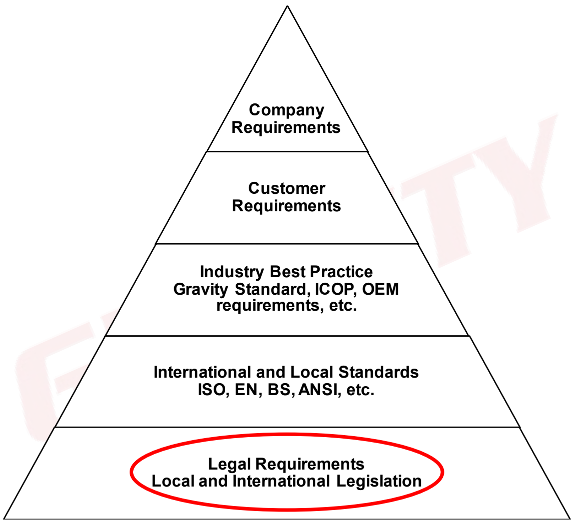
The following information must be captured on the COF:

- The employee's full name and surname.
- The employee's ID number.
- The date of the examination.
- The date of expiry (the COF must not be older than one year).
- The result of the examination (fit or unfit as per job specification).
- The signature of the OHP that performed the medical evaluation.

The FPP must include the correct procedure for medical evaluations relevant to the site and scope of work. It is the duty of the FPP implementer to ensure that this procedure is followed by everyone on the project and that the COFs, as well as any other required proof of this implementation, are available on request.

It is also recommended that a COF register is kept and managed by an appointed responsible person (HSE representative, safety officer, site supervisor, etc.), indicating the COF expiry date as well as the date of the next examination.

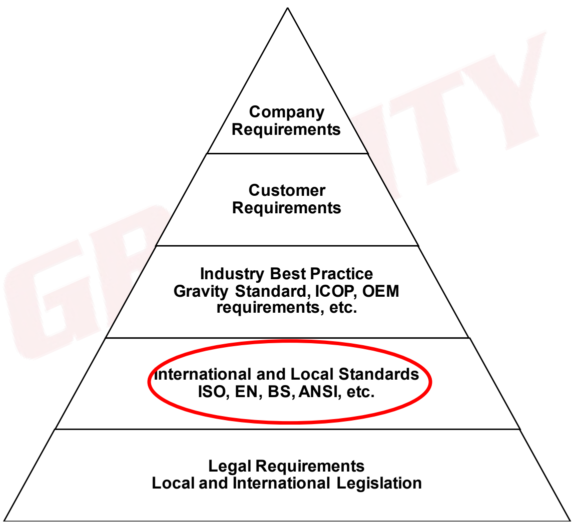
Legal Requirements



Construction Regulation 10(2)(b)
The processes for the evaluation of the employee's medical fitness necessary to work at a fall risk position, and the records thereof.

Construction Regulation 7(8)
A contractor must ensure that all his/her employees have a valid medical COF specific to the construction work to be performed and issued by an OHP in the form of Annexure 3.

International and Local Standards

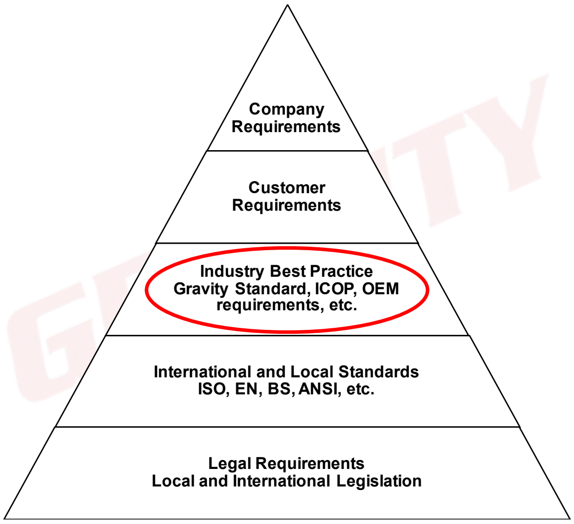


Our next building block is international standards. International standards are very important, especially when working with customers that operate in more than one country.

Standards like ISO and ANSI refer to periodic medical examinations to determine fitness to work at height.

In most cases, the frequency of these examinations are determined by the industry or the customer.

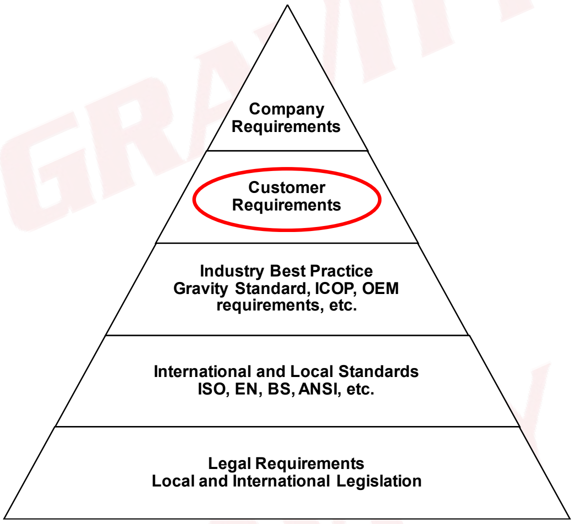
Industry Best Practice



Following international standards, we look at what the rest of the industry is doing.

The question of why a medical must be done annually is answered under industry best practice: Given the nature of working at height, it is an inherently dangerous activity. As such, the industry has settled on annual medical evaluations. But it does not stop there. Workers need to communicate any change in medical condition to the supervisor immediately!

Customer Requirements



Then, we consider customer requirements.

A customer may require additional criteria regarding medical fitness and more frequent medical evaluations than the industry norm.

For example, mines require entry and exit medicals, even if a worker will only be there for a few days. They also require that you receive a medical from their preferred supplier – this means that even if you had a medical examination done yesterday, they might still require a new one today.

Let's look at an example of what is required by the construction regulations.

ANNEXURE 3
OCCUPATIONAL HEALTH AND SAFETY ACT, 85 OF 1993 CONSTRUCTION REGULATIONS, 2014
MEDICAL CERTIFICATE OF FITNESS

Name of Employee ____________________ ID Number ____________________ Co Number __________

*Occupation e.g. General Worker, Welder, Bricklayer, Steel fixer, Mobile Crane Operator, etc.	*Possible Exposures e.g. noise, heat, fall risk, confined space, etc.	*Job Specific Requirements e.g. Operating Mobile Crane, Digging Trenches, Erecting Form work & Support work, etc.	*Protective Equipment e.g. Dust Respirator (light duty), Welding Gloves, etc.

***The employer to complete the information in the spaces marked with an * before sending the Employee for a medical examination**

Declaration by the Medical Examiner

I certify that I have, by examination and testing, using the above criteria specified by the employer, satisfied myself that the above mentioned employee is fit to perform the duties as described by the employer in the matrix above.

Occupational Medicine Practitioner / Occupational Health Nursing Practitioner: (Please Print Name) ______________________________

Signature ____________________ Practice Number ______________ Date ______________

Address ____

Let's look at an example of what is required by an industry best practice standard. This still meets the requirement of what should be on a COF as a minimum.

Certificate No:

REPUBLIC OF KENYA

SEAFARER MEDICAL CERTIFICATE

Issued under Regulation 6 of the Merchant Shipping (Seafarer Medical and Examinations) Regulations, 2016

SEAFARER INFORMATION

Last name:	First name:	Middle name(s):	
Nationality:	Passport No:	Date of Birth: __ /__ /____	Discharge book No:
Gender: ☐ Male ☐ Female	Rank/Job:	Department:	

This is to certify that above named seafarer has been examined in accordance with the Seafarers' medical fitness standards and certification requirements established in accordance with the provisions of the STCW Convention,1978 as amended, regulation I/9 and Maritime Labour Convention 2006, regulation1.2 found to be fit for sea service, subject to any limitations indicated

DECLARATION OF THE RECOGNIZED MEDICAL PRACTITIONER

	Yes	No
1.Confirmation that identification documents were checked at the point of examination	☐	☐
2. Hearing meets the standards in STCW Code Section A-I/9? Date of test *(dd/mm/yyyy)*: ___/___/_____	☐	☐
3. Unaided hearing satisfactory?	☐	☐
4. Visual acuity meets standards in section A-I/9? Visual aids (if worn): ☐Spectacles ☐Contact lenses ☐None	☐	☐
5. Colour vision meets standards in section A-I/9? Date of last color vision test: ______/____/______	☐	☐
6. Fit for look-out duties? *(Deck and Engine Dept. only)*	☐	☐
7. Limitations or restrictions on fitness?	☐	☐

If "Yes", specify limitations or restrictions: Examination form No:

	Yes	No
8. Is the seafarer free from any medical condition likely to be aggravated by service at sea or to render the seafarer unfit for such service or to endanger the health of other persons on board?	☐	☐

Date of Issue*(dd/mm/yyyy)*: ______/____/______ Date of Expiry* *(dd/mm/yyyy)*:____/____/______

RECOGNIZED MEDICAL PRACTITIONER	**REGISTRAR OF KENYAN SEAFARERS**
Sign: _ _ _ _ _ _ _ _ _ _ _ _ _ _	Sign: _ _ _ _ _ _ _ _ _ _ _ _ _ _
Name (print): _ _ _ _ _ _ _ _ _ _ _ _ _ _	Name (print): _ _ _ _ _ _ _ _ _ _ _ _ _ _
Place of examination: _ _ _ _ _ _ _ _ _ _ _ _ _ _	Place of issue: _ _ _ _ _ _ _ _ _ _ _ _ _ _
Seal/Stamp: ____________________	Seal/Stamp: ____________________

SEAFARER'S DECLARATION: I hereby confirm that I have been informed about the content of this certificate and my right to appeal in accordance with the Merchant Shipping (Seafarer Medical and Eyesight) Regulations, 2016.

Signature of the seafarer: ____________________ **Valid for a maximum period of two years unless the seafarer is under the age of 18, in which case the maximum period of validity shall be one year.*

Activity 5

Write a medical evaluation procedure summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; **when** will it be implemented; and how will they **prove** it has been implemented?

G. TRAINING PROGRAMME AND TRAINING MATRIX

1. Training Programme

To ensure that all workers onsite are aware of the correct methodology to perform a given task safely, workers must undergo training.

The required training is not limited to fall arrest or rope access training but includes all training required for a worker to perform their task effectively, competently and, above all, safely.

The FPP will include a training matrix which will stipulate the training required as per job specification, as well as when and how often the training must be received.

Typical work-at-height training includes, but is not limited to:
- Basic Fall Arrest
- Fall Arrest and Basic Rescue
- Fall Protection Plan Development
- Implementing a Fall Protection Plan
- Advanced Fall Arrest Rescue
- Rope Access Basic Operative
- Rope Access Intermediate Operative
- Rope Access Advanced Operative (Supervisor)
- Rope Rigging
- Mechanical Lifting
- Climbing Equipment Inspection and Management
- Portable User/Portable Ladder and Pole Climbing

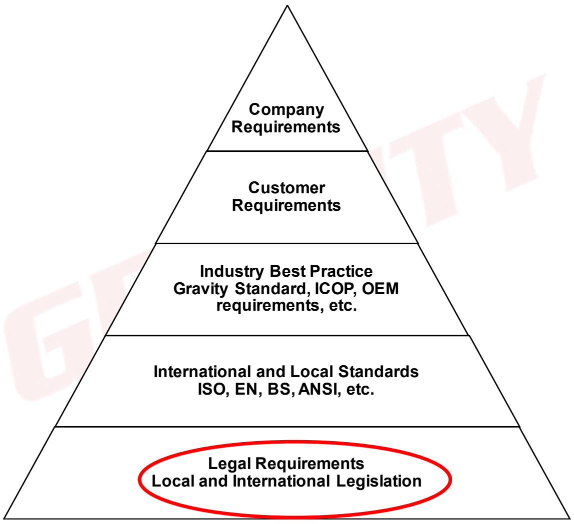
Apart from training in the specific access method, all workers must also receive training on the FPP – this is referred to as FPP induction, and includes an induction on the:
- risk assessment,
- emergency procedures and
- safe operating procedures.

The supervisor will receive FPP implementation training as they will be the person responsible for the implementation of the FPP.

The FPP will include the frequency at which training must take place (some training might be required every three years; some might be required more than once a day). It is the duty of the implementer (supervisor) to ensure that there is proof that the required training has been conducted at the required intervals as set forth by the FPP.

Proof of all training must be available on request. This may be given in the form of certificates, induction registers or attendance registers (where proof of attendance and not competency is required).

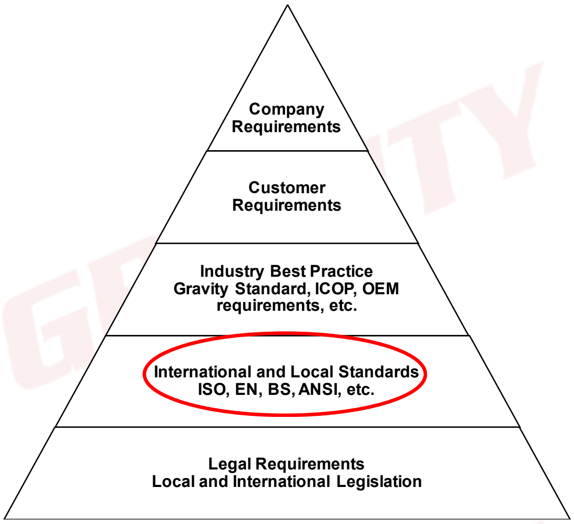
Legal Requirements



Construction Regulation 10(2)(c)
A programme for the training of employees working from a fall risk position, and the records thereof.

Construction Regulation 9(3)
A contractor must ensure that all employees under his/her control are informed, instructed and trained by a competent person regarding any hazard and the related work procedures and/or control measures before any work commences, and, thereafter, at the times determined in the risk assessment monitoring and review plan of the relevant site.

International and Local Standards



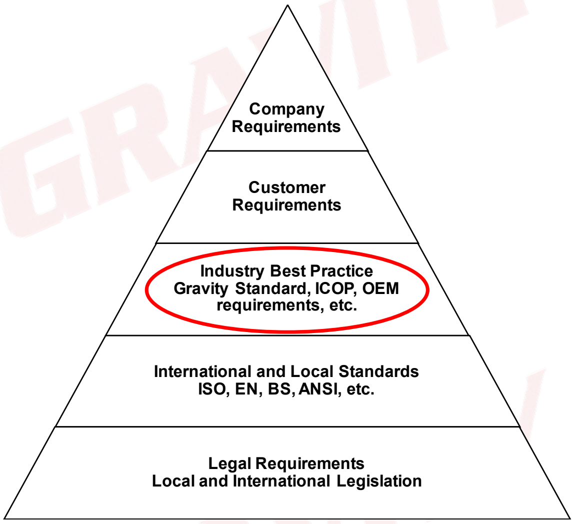
Our next building block is international standards. International standards are very important, especially when working with customers that operate in more than one country.

ANSI A10.48 and Z490 make specific references to the requirements of training programmes, including training records and training elements.

Depending on the environment where you plan on operating, it is important to consider these requirements:

- Competent and experienced training providers must be used.
- Training must be relevant to the scope of work.
- Theoretical and practical components must be incorporated.
- Records of training must be kept.

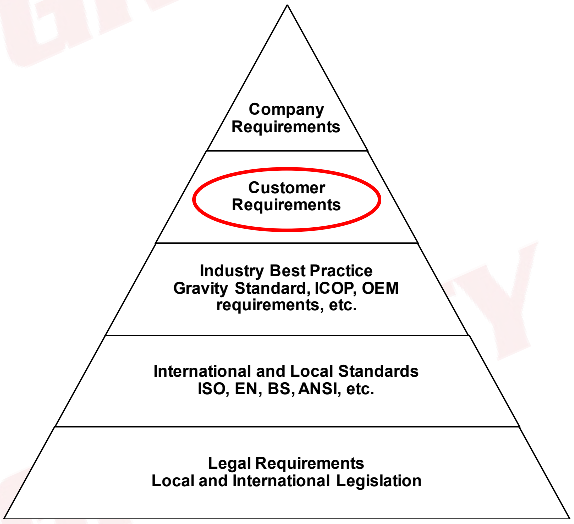
Industry Best Practice



Following international standards, we look at what the rest of the industry is doing.

With regard to the frequency of training, the period before a refresher is required will be determined by the severity of the hazard a worker is exposed to. In terms of work at height, it has a high likelihood of fatality, and as such, the industry has settled on retraining or refresher training at least every three years. This was also incorporated into ISO 22846-2.

Customer Requirements



Then, we look at customer requirements.

A customer may require additional training, which may not be a legal requirement in your country, depending on the scope of work.

A great example of this is solar training, that is a requirement by a specific customer and only marketed to their contractors. However, companies are encouraged to pursue this training as it also includes elements that are beneficial to other work environments.

Activity 6.1

Write a training programme summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; **when** will it be implemented; and how will they **prove** it has been implemented?

2. Training Matrix

To organise our training into an easily understandable table, we can make use of a training matrix. We read our designation and match it with the training required.

For example:
The FPP developer will require fall arrest and basic rescue training, as well as FPP developer training. Because we stated in our policies and procedures that the supervisor will be responsible for implementing our FPP, all supervisors will require FPP implementation training.

When compiling your training matrix, be sure to cover the full scope identified in your risk assessment. For example, if you are working on a telecoms site, RFA becomes a requirement for all employees on that site.

Legend:

- FPP Induction – Fall Protection Plan Induction
- FPP Implementation – Fall Protection Plan Implementation
- BFA – Basic Fall Arrest
- FABR – Fall Arrest and Basic Rescue
- PLU – Portable Ladder User
- PLPC – Portable Ladder and Pole Climbing
- AFR – Advanced Fall Arrest Rescue
- FPPD - Fall Protection Plan Developer.
- RR – Rope Rigging
- ML – Mechanical Lifting
- CEIM – Climbing Equipment Inspection and Management
- RA L1 – Rope Access Level 1
- RA L2 –Rope Access Level 2
- RA L3 – Rope Access Level 3
- FA – First Aid
- RFA – Radio Frequency Awareness

Fall Risk Training Matrix							
Training requirements per designation from SETA/QCTO-approved training provider (trainer and assessor to be a "competent person")							
No.	**Designation**	**FPP Induction**	**FPP Implementation**	**BFA**	**FABR**	**FPP**	**RFA**
1.	FPP Developer				Yes	Yes	

Fall Risk Training Matrix							
Training requirements per designation from SETA/QCTO-approved training provider (trainer and assessor to be a "competent person")							
No.	**Designation**	**FPP Induction**	**FPP Implementation**	**FABR**	**RR**	**FA**	**RFA**
1.	Construction Supervisor		Yes	Yes	Yes	Yes	Yes

Fall Risk Training Matrix							
Training requirements per designation from SETA/QCTO-approved training provider (trainer and assessor to be a "competent person")							
No.	**Designation**	**FPP Induction**	**FPP Implementation**	**FABR**	**CEIM**	**FA**	**Other**
1.	Equipement Controller	Yes		Yes	Yes	Yes	OEM Training

Note:
Anyone that works in an area where there may be a risk of exposure to radio frequency radiation, must receive RFA training before work starts.

Fall Risk Training Matrix

Training requirements per designation from SETA/QCTO-approved training provider (trainer and assessor to be a "competent person")

No.	Designation	FPP Ind	FPP Imp	BFA	FABR	PLU	PLPC	AFR	FPPD	RR	ML	CEIM	RA L1	RA L2	RA L3	FA	RFA	Other
1	CEO	Yes																
2	Assistant to the CEO	Yes																
3	FPP Developer				Yes				Yes									
4	Equipment Controller	Yes			Yes							Yes						OEM Specific Trainig
5	Construction Supervisor		Yes													Yes		
6	Fall Arrest Technician	Yes		Yes														
7	Basic Fall Arrest Rescuer	Yes			Yes											Yes		
8	Advanced Fall Arrest Rescuer	Yes						Yes					Yes			Yes		
9	Rigging Supervisor (<100 kg)		Yes	Yes						Yes								
10	Basic Rigger (<100 kg)	Yes		Yes						Yes								
11	Rigging Supervisor (<500 kg)		Yes							Yes	Yes							
12	Rigger (<500 kg)	Yes								Yes	Yes							
13	Ladder User	Yes		Yes		Yes												
14	Pole Climber	Yes		Yes			Yes											
15	Telecoms Operative	Yes															Yes	
16	RA Basic Operative	Yes											Yes					
17	RA Intermediate Operative	Yes												Yes		Yes		
18	RA Advanced Operative		Yes												Yes	Yes		

Activity 6.2

Complete the training matrix for the minimum required training per designation for the fall arrest site (Site 1) and the rope access site (Site 2).

Training Matrix		
No.	Designation	Minimum Training Requirements
General		
1.	FPP Developer	
2.	Equipment Controller	
Site 1 – Lattice Tower and Rigging (Telecoms)		
Number of workers onsite:		
1.	Site Supervisor	
2.	Technician	
3.	Rescuer	
Site 2 – Rope Access (Simple Site)		
Number of workers onsite:		
1.	Site Supervisor	
2.	Technician	
3.	Rescuer	

H. EQUIPMENT MANAGEMENT PROCEDURE

Referring to the risk assessment, another cause of people, tools or equipment falling is equipment failure. To prevent equipment failure, there must be a procedure in place for the inspection, testing and maintenance of equipment. The inspection, testing and maintenance of equipment must be done by a person who is competent not only in CEIM but also in fall arrest, to be able to ascertain the suitability and safety of the equipment for a given task.

All equipment used onsite must be inspected at intervals specified by the manufacturer or the relevant standards. **Always follow OEM requirements regarding equipment.**

According to ISO 22846, work-at-height equipment must be inspected before, during and after each use. It must also be routinely inspected every six months, and intermittently if it has been used in exceptional events.

The relevant inspection criteria, storage requirements and replacement procedures of equipment is given here for the benefit of the FPP developer. It provides a basis for the FPP developer to be able to make a judgement on the sufficiency of the equipment management.

All requirements, policies, procedures and responsibilities for equipment management must be documented and recorded in the FPP, and it is the duty of the FPP implementer to ensure implementation. It is important to note that, although some regulations may call for an equipment testing procedure, no PPE may be used after it has been subjected to a load test, drop test or any form of destructive testing. Should companies wish to test equipment, refer all requirements to the OEM and record all relevant procedures in the FPP.

Inspection Criteria

Software	Hardware
	
• Condition of stitching • Discolouration • Wear and tear • Cuts • Burns • Date of manufacture • Condition of core (rope inspections)	• Rust • Deformation • Working action • Movement • Spring action • Wear and tear

1. Equipment Storage

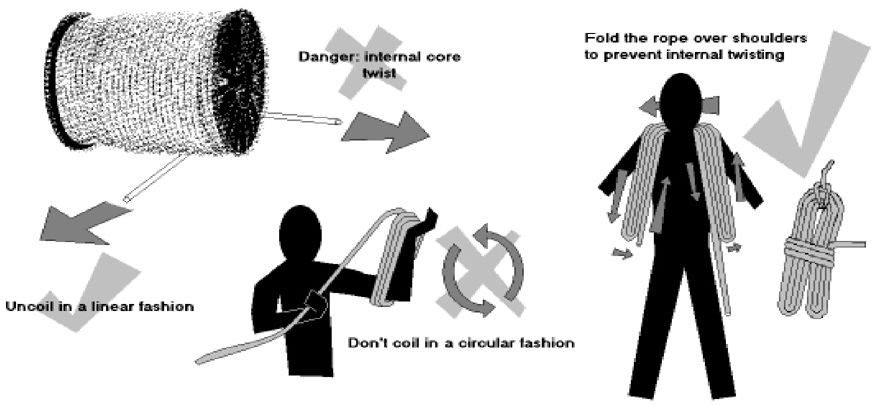
Equipment must be stored correctly and according to the manufacturer's specifications to prevent damage to equipment when not in use.

Things to consider when storing equipment in a SAFE way:

- **S**ecure location. This will prevent unauthorised personnel entering the equipment storage area and potentially damaging equipment.
- **A**way from direct sunlight, in a cool, dry place. This will help prevent mildew on software and rust on hardware.
- **F**ree of chemicals. This will reduce the chance of chemical damage to equipment.
- **E**nsure that there are no infestations (e.g. rats or moths) which can damage software. Have a system in place to prevent infestations.

2. Rope Care

- Do not use a rope for anything other than work at height.
- Do not run a rope over sharp edges. Use edge protection when there are sharp edges.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any other equipment being used.



3. Reporting Suspect Equipment

Equipment that is found to be damaged or too old for use must be reported to the person responsible for the site (i.e. the site supervisor, safety officer, etc.).

Damaged or expired equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.

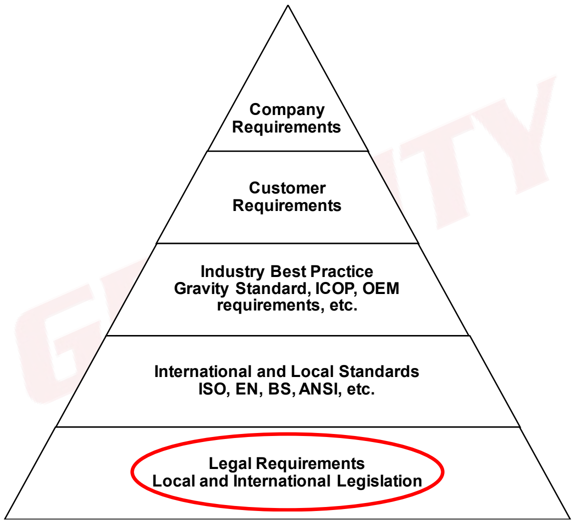
Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer to do so. **No contractor may repair PPE.**

All the procedures pertaining to equipment will be included in the FPP. It is the duty of the FPP implementer to ensure that these equipment procedures are complied with onsite. These procedures include requesting inspection records, rejection notices, requisition forms, etc.

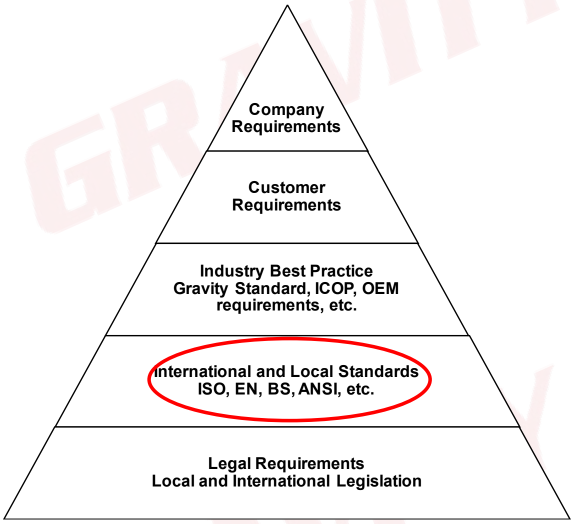
Suspect Equipment Found → **Report to Supervisor** → **Isolate Suspect Equipment** → **Hand Equipment to Equipment Controller** → **Equipment Rejected: Destroy Equipment; Issue New Equipment** / **Equipment Passed: Continue as Normal**

Legal Requirements



Construction Regulation 10(2)(d)
The procedure addressing the inspection, testing and maintenance of all fall protection equipment.

International and Local Standards



ISO 22846-2
All rope access equipment should be suitable for the intended purpose and selected in accordance with defined criteria, which should be set out in the company's management system.

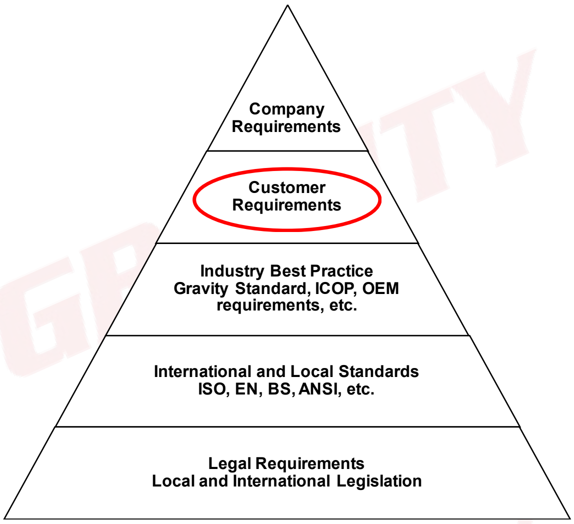
The specific details of any worksite or work task should be considered when selecting equipment.

Industry Best Practice

Considering the frequency at which equipment must be inspected, we refer to ISO for the baseline. According to ISO, equipment must be inspected every six months.

However, some customers have stricter requirements, stating that equipment must be inspected by a competent person every three months.

Customer Requirements



A customer may require that a competent person perform more frequent inspections on equipment, depending on the equipment's frequency of use and severity of exposure in their scope of work.

This may be motivated by their commitment to safety or their experience in their environments.

Activity 7.1

Write an equipment management procedure summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; **when** will this be implemented; and how will they **prove** it has been implemented?

Activity 7.2

Write an SOP for what to do when you find suspect equipment onsite.

Complete the practice activities as preparation for your assignment!

Remember:
SOPs have a **scope, purpose** and **methodology**.

I. RESCUE PLANNING

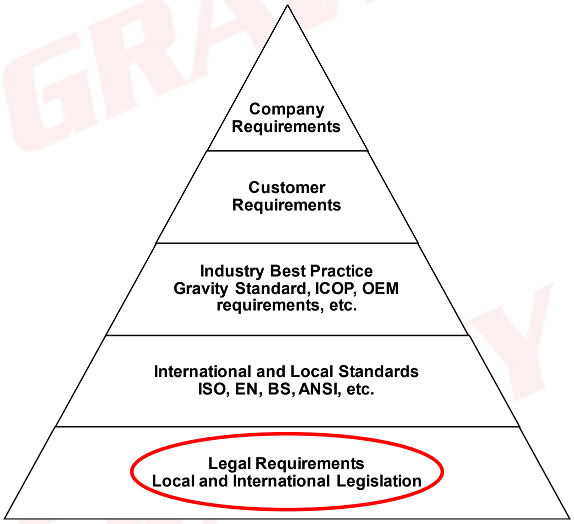
Before any work may start onsite, you must consider the worst-case scenarios, which includes all emergency procedures as described in the HSE file as well as the site-specific fall rescue plan.

It is the duty of the FPP implementer to ensure that:

- all workers onsite are aware of the rescue to be conducted (should a rescue need to be performed),
- the rescue plan is applicable to the site and
- there is a competent rescuer and a first aider onsite.

The FPP will include a rescue plan. The rescue plan will be site- and task-specific and will be written to the competencies of the rescuers onsite. It is the duty of the FPP implementer to ensure that the necessary equipment and personnel are present onsite (should a rescue need to be performed) and that all workers are aware of the relevant emergency procedures. Proof that the workers are aware of these procedures and that all the necessary equipment and personnel are available onsite should be given in the form of induction training registers, pre-task risk assessments and rescue training certificates.

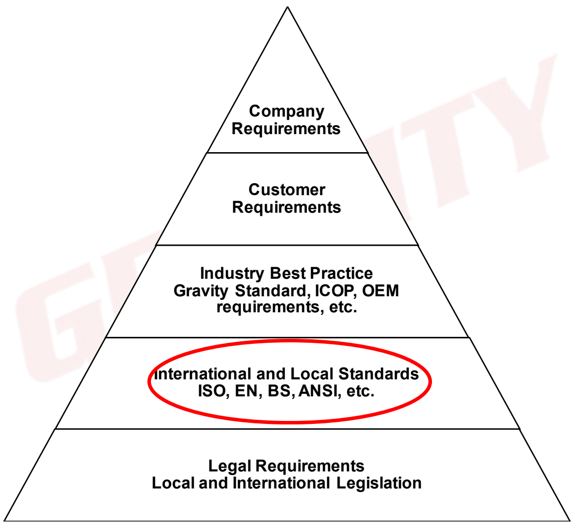
Legal Requirements



Construction Regulation 10(2)(e)
A rescue plan detailing the necessary procedure, personnel and suitable equipment required to effect a rescue of a person in the event of a fall incident, to ensure that the rescue procedure is implemented immediately following the incident.

General Safety Regulation 3(1)
An employer shall take all reasonable steps that are necessary under the circumstances to ensure that persons at work receive prompt first aid treatment in case of injury or emergency.

International and Local Standards

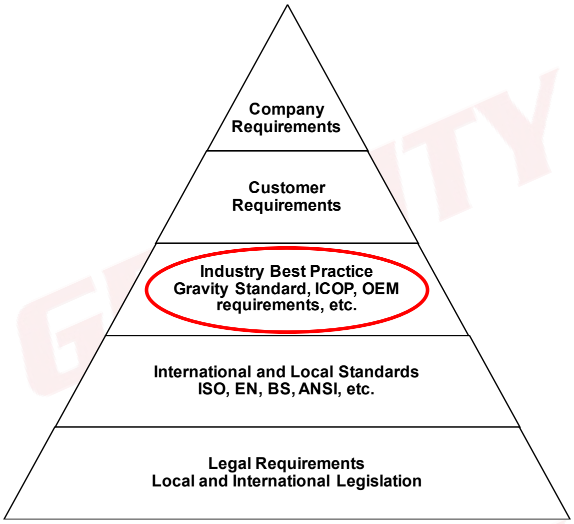


Standards relating to rescue requirements are quite prominent in the work-at-height industry.

ISO is specific on rescue requirements for rope access. ANSI refers to a number of fall arrest rescues that are popular in Africa.

Overall, these standards have one common denominator – a rescue should be specific to the site where the work is conducted.

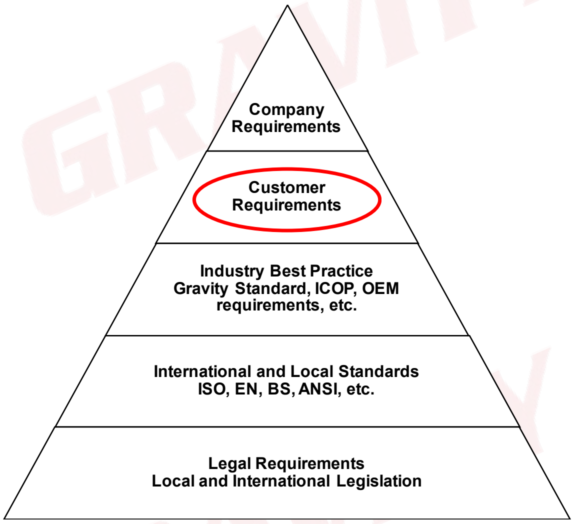
Industry Best Practice



Depending on the industry, it is becoming more common to train workers in fall arrest and basic rescue to enable any person working at height to perform a basic rescue.

As such, advanced rescue teams are only on standby where a basic rescue will not be effective. The advanced rescue teams are competent in advanced fall arrest rescue and higher-level rope access training.

Customer Requirements



Customers may have specific requirements regarding rescue, depending on their site (as they would have knowledge about what type of rescue would be the most effective) or depending on their commitment to safety. One example of this is a customer that requires **all** persons working from a fall risk position to have basic rescue training, instead of the regular requirement of minimum two per team.

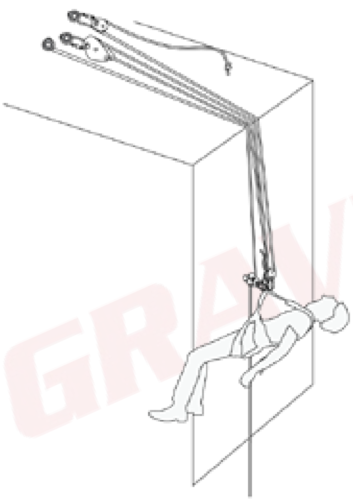
1. Site-Specific Rescue

The reason for stating "site-specific rescue" is because a rescue is not "one size fits all". This is a very important consideration to make when planning work at height, because your team may require additional training and/or different equipment for different sites. Consider the following scenarios:

In the event that the point of safety is below a casualty, you would have to make use of a **lowering rescue** to get your casualty to safety.

There are many options in terms of lowering rescues, of which the most common one is the **ratchet rescue**. This rescue makes use of a ratchet to lift the casualy a small amount in order to disconnect the deployed shock absorbers, and then it makes use of a descender to lower the casualty to the point of safety.

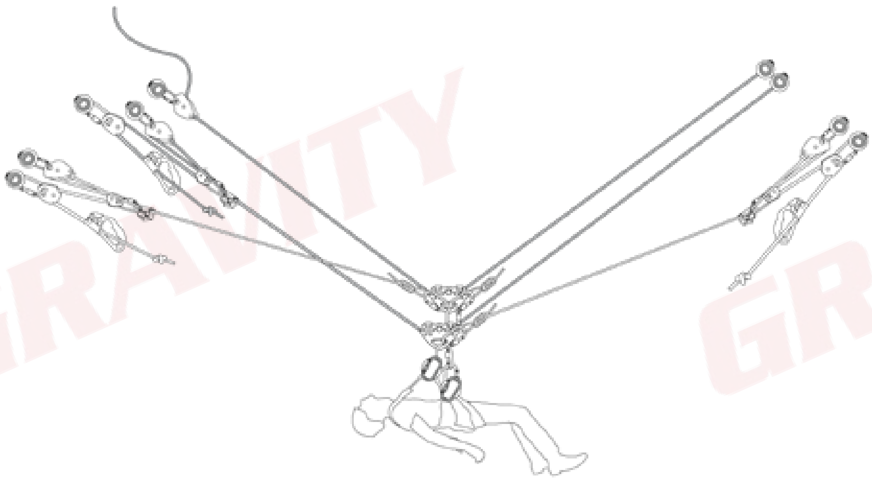
At Gravity, this is the rescue taught when you attend our Fall Arrest and Basic Rescue course.



In the event that the point of safety is above a casualty, you would have to make use of a **lifting rescue** to get your casualty to safety.

Lifting rescues often require the use of a mechanical advantage system to lift a casualty, using pulleys and progress capture devices.

At Gravity, this is one of the rescues taught when you attend our Advanced Fall Arrest Rescue course.



In the event that the point of safety is to the side of a casualty, you would have to make use of a **diagonal rescue** to get your casualty to safety.

This is one of the more advanced rescues you can use and is dependent on rope access methodologies. This is also commonly known as a **cableway rescue** and requires a main line and a backup line.

At Gravity, this is one of the rescues taught when you attend our Rope Access Level 3 course.

You also need to consider that some sites may require you to make use of rig-for-rescue systems, because the traditional rescues may not be possible. What this means is that a rescuer might not be able to install a rescue system after a casualty has fallen, due to access restrictions like not being able to climb past a casualty.

Rig-for-rescue is based on the principle that the rescue system is in place or rigged before the work starts, and if a person then falls, they can be rescued immediately. Some good examples of this are:

- temporary horizontal lifelines with a descender installed,
- monopole climbing methodology where a descender is used to belay the climber and
- lifelines rigged through descenders at the anchor point.

2. Methodologies for Different Rescues

i. Ratchet Rescue Procedure

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to the casualty's top D-ring.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between the casualty and the descender.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

When performing a rescue, remember that communication is of utmost importance and **must be maintained at all times**.

To save time when attempting a rescue, as well as to reduce the risk of errors occurring and/or dropping items when at height, it is always better to pre-rig the rescue system when on the ground before climbing up to the casualty.

ii. 3:1 Rescue Procedure

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to the casualty's top D-ring.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the descender to the structure.
7. Connect the chest ascender and pulley to the rope.
8. Connect the rescue pulley to the structure.
9. Ensure that all slack is removed between the casualty and the descender.
10. Lift the casualty.
11. Remove the casualty's lanyard.
12. Remove the 3:1 system.
13. Apply extra friction.
14. Lower the casualty.

When performing a rescue, remember that communication is of utmost importance and **must be maintained at all times**.

To save time when attempting a rescue, as well as to reduce the risk of errors occurring and/or dropping items when at height, it is always better to rig the rescue system when on the ground before climbing up to the casualty.

iii. Same-Rope Rescue from a Descender Procedure

1. Connect your backup device on the casualty's main line.
2. Ascend until you are the same level as the casualty.
3. Connect your cowtail's backup device above the casualty's descender.
4. Connect your backup lanyard to the backup device above the casualty's descender, and remove your cowtail.
5. Connect the casualty's cowtail to their backup device and remove their backup lanyard.
6. Ascend until you are higher than the casualty.
7. Perform a changeover.
8. Connect your extra cowtail to the casualty's **bottom D-ring**.
9. Connect the rescue sling to **your descender's karabiner** and the **casualty's top D-ring**.
10. By controlling the casualty's descender, slowly lower the casualty until the descender becomes slack.
11. Remove the casualty's descender from the casualty's harness, **but not from the rope**.
12. Connect the casualty's descender to your harness; **ensure that the gates of both karabiners are facing downward and towards you**; remove all the slack and lock the descender.
13. Remove your backup device.
14. Use the karabiner connecting the casualty's backup lanyard to their harness and connect this karabiner to the slack ends of **both** ropes.
15. While keeping tension on the slack ends of the ropes running through the extra karabiner, descend in a controlled fashion.
16. Open both karabiners and remove your **bottom D-ring**.

Activity 8.1

Write a rescue plan summary for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when** will this be done? And then, **who** will do **what**; **when** will it be implemented; and how will they **prove** that it has been implemented?

Activity 8.2

Develop a site-specific rescue SOP for the site you selected in Activity 3. Include a list of the equipment (with the EN standards) that will be required for the rescue.

Complete the practice activities as preparation for your assignment!

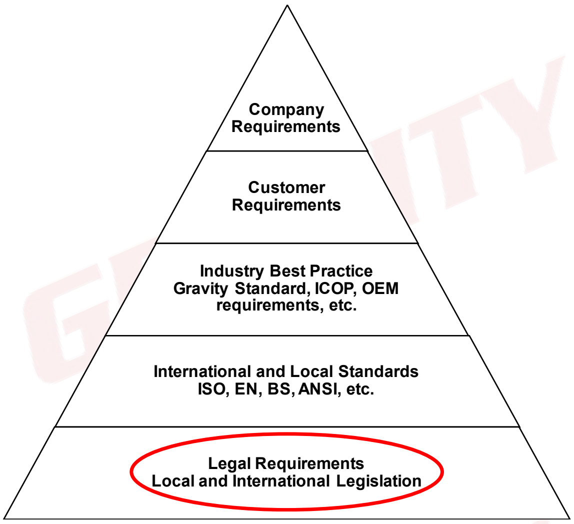
Remember:
SOPs have a **scope, purpose** and **methodology.**

J. MONITOR AND REVIEW

An FPP must be reviewed to ensure that it is still up to date and still manages all fall risks onsite. The FPP implementer (site supervisor) need not review the plan themself. However, it is the duty of the implementer to monitor the plan and inform the FPP developer should the plan become insufficient or outdated, or in case of the following:

- if a change in legislation has occurred,
- if a change in the scope of work has occurred,
- after an incident has occurred,
- after one year has passed since the plan was written or last reviewed, or
- should a customer request it.

Legal Requirements

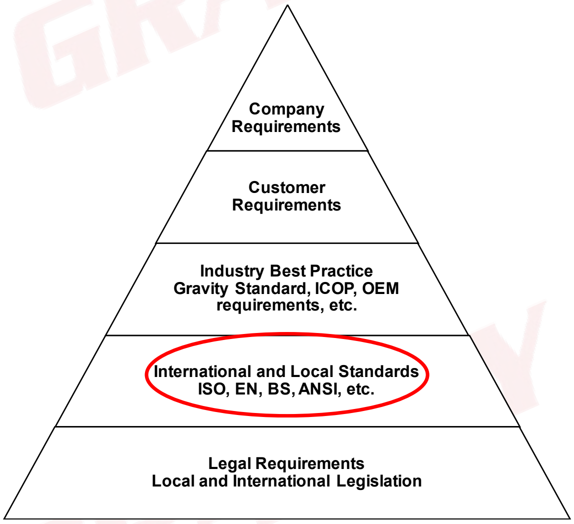


Construction Regulation 10(1)(b) and (c)

b. Ensure that the FPP contemplated in paragraph (a) is implemented, amended where and when necessary and maintained as required; and

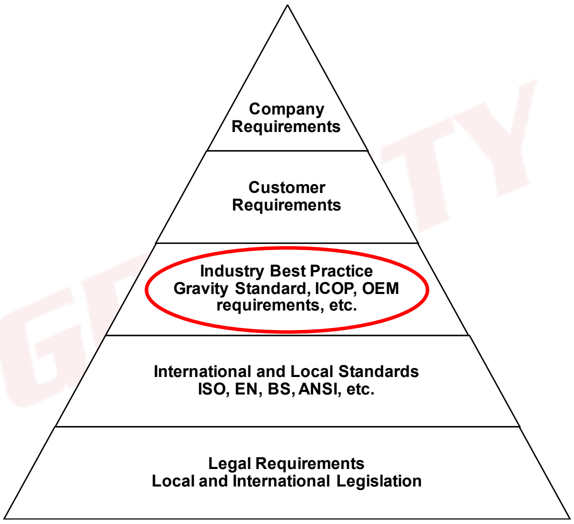
c. take steps to ensure continued adherence to the FPP.

International and Local Standards



To monitor and review the FPP is frequently mentioned in all international standards. The main purpose of these actions is to ensure continuous improvement throughout the application of the plan. As such, all of the standards state that the FPP must be monitored and reviewed, though there are slight variations on the conditions under which this must be done and how frequenctly it must be done.

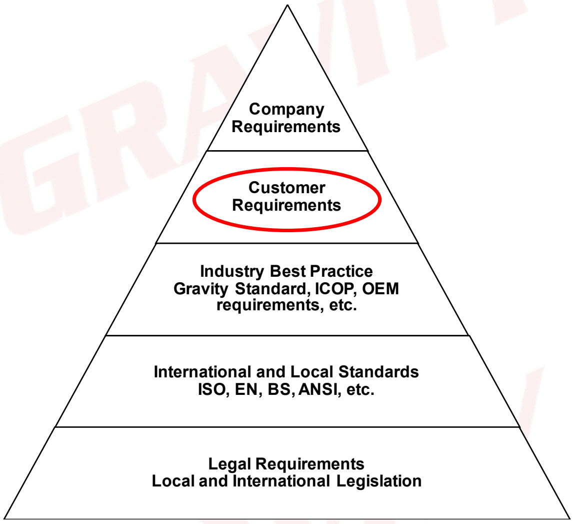
Industry Best Practice



Monitoring implementation of FPPs and revising such plans are critical to ensuring the success of these plans.

In the industry, it is common practice to review these plans frequently (no more than annually) to ensure that they remain relevant and effective. Beyond that, an FPP must also be reviewed in the event of an incident or when a change in the scope of work occurs.

Customer Requirements



It is important to consider that, above and beyond legal, international and industry requirements, a customer may also require that you have certain additional elements in place to prevent persons and/or objects from falling.

These requirements must be considered when developing and reviewing an FPP. A customer may expect your company to do more than the standards require, but they cannot expect you to do less .

Activity 9

Write a monitor and review procedure for your FPP.

Complete the practice activities as preparation for your assignment!

Remember:
Who will do **what**, **how**, for what **logical** and **legal** reason, and **when (x5)** will this be done?

K. ADDITIONAL CONTENT

You have now completed the basic minimum requirements for your FPP. However, remember to add specific SOPs to your scope of work.

This can include (but is not limited to) SOPs for:

- Ladder work
- Scaffolding
- Fall arrest
- Rope access
- Rigging
- Mechanical lifting

This can also include a procedure for incident reporting, though this typically forms part of your greater HSE management system. Any and all incidents must be reported in accordance with the relevant regulations using the relevant document templates provided.

Should there occur any operation that falls outside the relevant accepted standards (e.g. operations outside the OEM requirements), work must be stopped and the transgression reported to the construction manager/risk assessor/safety officer/ HSE representative.

Any and all reported transgressions must be documented and recorded with as much detail as possible. This will aid the investigation, should there be one, as well as allow the cause of such transgressions to be identified and removed. Workers must report any and all non-conformances in order for an investigation to be conducted and so that a safe worksite may be established and maintained.

Should the FPP implementer be unsure of the course of action, they must contact the FPP developer in order to determine the best way forward. All incidents must be recorded and investigated in order to determine the source/cause of the incident so that it may be prevented in future and to help maintain a worksite that is as safe as possible.

Activity 10

Write an SOP for the fitment and safe use of a shock absorber (while moving up and down a structure).

Complete the practice activities as preparation for your assignment!

Remember:
SOPs have a **scope, purpose** and **methodology.**

THE WAY FORWARD

Congratulations on completing Gravity Training's Fall Protection Plan Development course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.